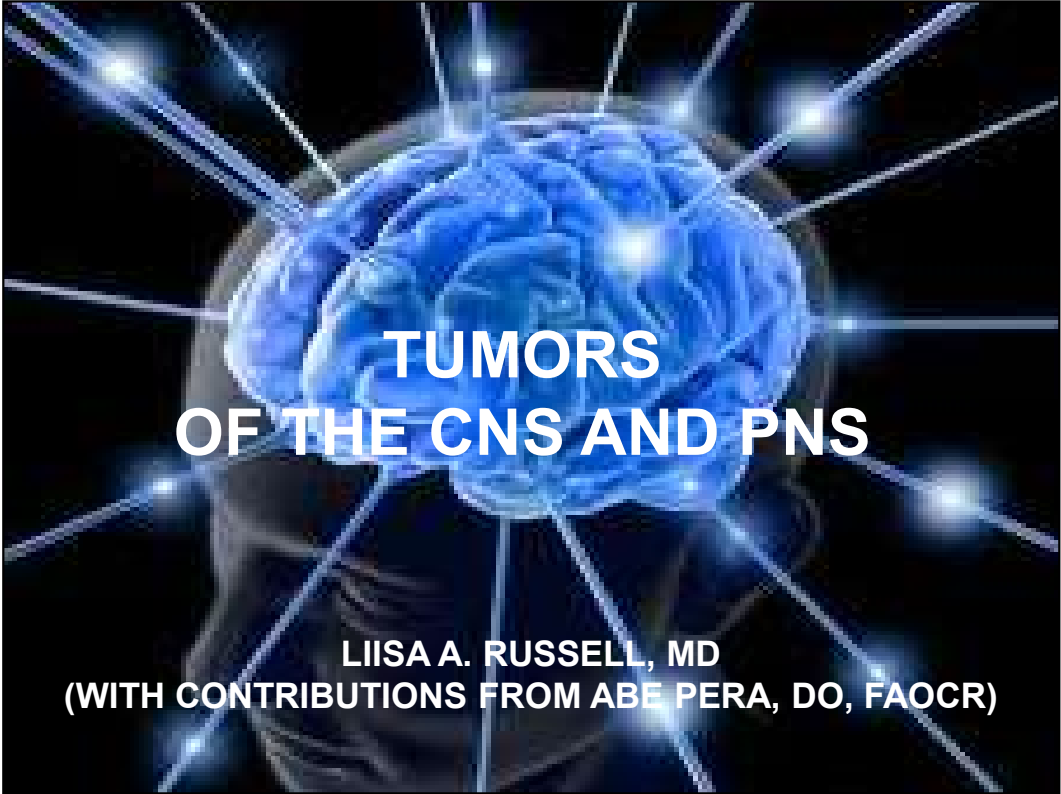




TUMORS OF THE CNS AND PNS

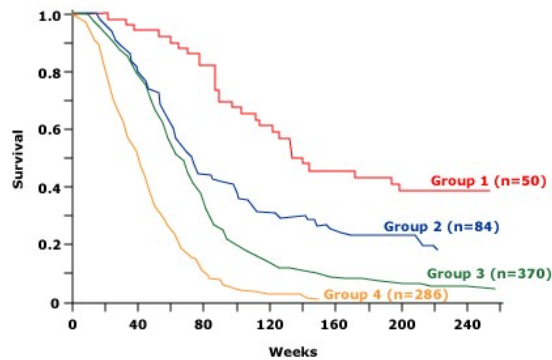
**LIISA A. RUSSELL, MD
(WITH CONTRIBUTIONS FROM ABE PERA, DO, FAOCR)**

LEARNING OBJECTIVES

1. Know the common clinical presentation of CNS tumors
2. Know the common types of CNS tumors
3. Know the preponderance of types of CNS tumors presenting in various age groups
4. Know the classification and grading schemes for CNS tumors and their significance
5. Describe the characteristic clinical and morphologic features of gliomas
 - a. Astrocytomas:
 - Pilocytic astrocytoma
 - Fibrillary astrocytoma
 - Glioblastoma multiforme
 - b. Oligodendroglioma
 - c. Ependymoma
6. Describe the characteristic clinical and morphologic features of PNETs (e.g. medulloblastoma)
7. List the neuronal tumors in the nervous system
8. Describe the characteristic clinical and morphologic features of germ cell tumors in the CNS
9. Describe the characteristic clinical and morphologic features of primary CNS lymphomas
10. Describe the characteristic clinical and morphologic features of meningiomas
11. Describe the general features of metastatic tumors to the brain
12. Know the inheritance pattern and list the most commonly occurring neoplasms in familial tumor syndromes
 - a. Type 1 and 2 neurofibromatosis - autosomal dominant
 - b. Tuberous sclerosis – autosomal dominant
 - c. von Hippel-Lindau – autosomal dominant
13. Describe the characteristic clinical and morphologic features of benign and malignant tumors and tumor-like lesions of the peripheral nerve
 - a. Schwannoma
 - b. Neurofibroma
 - c. Malignant peripheral nerve sheath tumor
 - d. Traumatic neuroma

EPIDEMIOLOGY OF CNS CANCERS

Survival for all patients, by risk group



Group 1: Age ≤ 40 , frontal tumor.
Group 2: Age ≤ 40 , other tumor sites.
Group 3: Age > 40 and < 65 ; KPS > 70 and gross or subtotal resection.
Group 4: Age ≥ 65 or age < 40 ; or KPS ≤ 70 ; or biopsy only.

MORTALITY AND PROGNOSTIC FACTORS

- CNS cancers are less common than lung or breast cancer, but cause more morbidity
- Overall survival not much changed over 50 years
- Younger age and lower pathologic grade and better Karnofsky score are favorable prognostic factors for primary brain tumors.

CLINICAL PRESENTATION

Initial symptoms in 565 patients with grade III or IV malignant glioma

Symptom	Grade III, percent	Grade IV, percent
Headache	53	57
Seizure	56	23
Memory loss	26	39
Motor weakness	25	36
Visual symptoms	23	21
Language deficit	22	36
Cognitive changes	22	39
Personality changes	11	27
Change in consciousness	11	18
Nausea and vomiting	8	15
Sensory deficit	5	12
Papilledema	5	5

Data from Glioma Outcomes Project, Chang, SM, et al, JAMA 2005; 293:557.

CLINICAL PRESENTATION

▣ SYMPTOMS AND SIGNS

- General or focal
- Causes
 - Local brain invasion
 - Local brain compression of adjacent structures
 - Increased intracranial pressure (ICP)

CLINICAL PRESENTATION - GENERAL

HEADACHES

Most common manifestation of CNS tumors

Dull and constant, occasionally throbbing

Severe headaches uncommon unless obstructive hydrocephalus or meningeal irritation

Early morning headache relatively common

ICP elevation classic triad of headache, nausea and papilledema

SEIZURES

More common with lower grade and primary tumors

May be generalized or focal (focal can become generalized)

Usually repetitive and stereotypical in a given patient

Differentiate from syncopal episode (from elevated ICP) which would require glucocorticoids to decrease pressure rather than anticonvulsants.

CLINICAL PRESENTATION - GENERAL

- ▣ NAUSEA AND VOMITING
 - **Due to increasing ICP** about the area postrema of the medulla (floor of the fourth ventricle)
 - Onset by abrupt change in position related to possible CNS tumor
 - Associated focal neurologic findings
- ▣ SYNCOPE
 - Elevated ICP can cause temporary cut-off of cerebral perfusion
 - ▣ Plateau waves are sustained pressure waves that normally occur within the brain and are caused by activities that transiently raise ICP such as Valsalva maneuver (cough, sneeze, vomiting) impeding JV flow.
 - ▣ In a patient with a tumor the baseline pressure may be elevated and any further increase causes syncope.
 - Important to differentiate from seizure disorders
 - Treatment is with glucocorticoids or surgery and not with anticonvulsants

CLINICAL PRESENTATION - GENERAL

- ▣ Memory problems
- ▣ Cognitive dysfunction
- ▣ Mood and/or personality changes
- ▣ Subtle symptom, e.g. low energy, fatigue, or sleepiness
- ▣ If symptoms are new, imaging is indicated

CLINICAL PRESENTATION

- ▣ FOCAL
 - Weakness
 - Sensory loss
 - Aphasia
 - Visual spatial dysfunction
 - ▣ Fundusoscopic exam:
papilledema



CLINICAL SYMPTOMS ASSOCIATED WITH BRAIN TUMORS

Mental status examination terminology

abulia: loss of initiative, willpower, or drive
acalculia: inability to calculate
agnosia: inability to recognize one or more classes of environmental stimuli, even though the necessary intellectual and perceptual functions are intact
agraphia: inability to write
alexia: inability to read for comprehension
amnesia: inability to retain new information
anomia: inability to name objects or think of words; in practice, often used as a synonym for dysnomia
anosognosia: inability to recognize one's own impairment
anterior aphasia: acquired language disorder in which verbal output is nonfluent
aphasia: literally, a complete loss of language function, but in practice, used as a synonym for dysphasia
aphemia: complete loss of the ability to speak, but retained comprehension and writing ability
apraxia: inability to perform a previously learned set of coordinated movements even though the necessary component skills (including intellect, language function, strength, coordination, and sensation) remain intact
Broca's aphasia: acquired language disorder characterized by nonfluent verbal output with omission of relational words (prepositions, conjunctions, articles, and minor modifiers) and abnormal prosody, impaired repetition, and relatively intact comprehension
conduction aphasia: acquired language disorder characterized by prominent impairment of repetition, relatively intact comprehension, and verbal output that is fluent but contains literal paraphasias
delirium: an acute confusional state characterized by clouded, reduced, or shifting attention, often associated with sensory misperception or disturbed thinking
dementia: acquired impairment of memory and at least one other cognitive function, without clouding of the sensorium or underlying psychiatric disease

10

Mental status exam terms

CLINICAL SYMPTOMS ASSOCIATED WITH BRAIN TUMORS

dementia: acquired impairment of memory and at least one other cognitive function, without clouding of the sensorium or underlying psychiatric disease
dysnomia: difficulty naming objects or finding the desired words
dysphasia: acquired disorder of language not due to generalized intellectual impairment or psychiatric disturbance
expressive aphasia: acquired language disorder in which verbal output is nonfluent
fluent: an adjective used to describe verbal output that is normal to excessive, easily produced, with normal phrase length (five or more words) and normal prosody
fluent aphasia: acquired language disorder in which verbal output is fluent
Gerstmann's syndrome: the constellation of (1) agraphia, (2) acalculia, (3) right-left confusion, and (4) finger agnosia; classically associated with lesions in the angular gyrus of the dominant hemisphere (but the subject of endless debate)
jargon: verbal output that contains so many literal paraphasias that the words are unrecognizable
nonfluent: an adjective used to describe verbal output that is sparse, with only one to four words per phrase
nonfluent aphasia: acquired language disorder in which verbal output is nonfluent
paraphasia: a substitution error in which the word produced is similar in sound or meaning to the intended word. A literal or phonemic paraphasia is a sound substitution error resulting in production of a word that is phonemically related to the intended word (eg, "greed" or "greeb" instead of "green"). A semantic or verbal paraphasia is a word substitution error in which the word produced is semantically related to the intended word (eg, "blue" instead of "green")
posterior aphasia: acquired language disorder in which comprehension is impaired
prosody: the rhythm and tempo of speech
prosopagnosia: inability to recognize faces
receptive aphasia: acquired language disorder in which comprehension is impaired
transcortical aphasia: acquired language disorder in which the ability to repeat is intact
Wernicke's aphasia: acquired language disorder characterized by markedly impaired comprehension and repetition, with verbal output that is fluent but contaminated by numerous paraphasias or, in severe cases, jargon

Mental Status exam terms

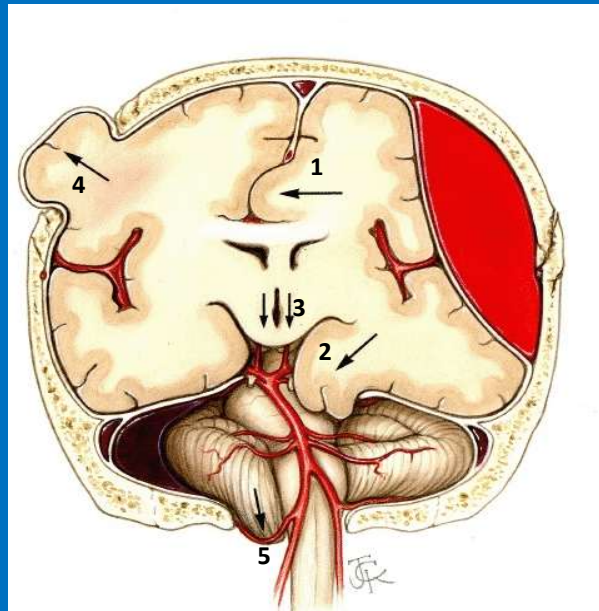
IMAGING OF BRAIN TUMORS

- ▣ The main purpose is to identify a lesion on initial evaluation after clinical evaluation
- ▣ Identify the extent of involvement
 - Size
 - Specific neurological territory and adjacent anatomy
 - Enhancement pattern
 - Evaluate proximity to vascular structures
- ▣ Identify complications
 - Hemorrhage, infarction, herniations

IMAGING OF BRAIN TUMORS

- ▣ HERNIATION OF BRAIN:
 - Brain is displaced from one compartment to another
 - ▣ Subfalcine
 - ▣ Unilateral descending transtentorial herniation (DTH)
 - Temporal lobe into tentorial hiatus
 - ▣ Bilateral DTH (Descending transtentorial herniation)
 - ▣ Ascending transtentorial herniation
 - Cerebellar lesion/mass

BRAIN HERNIATION

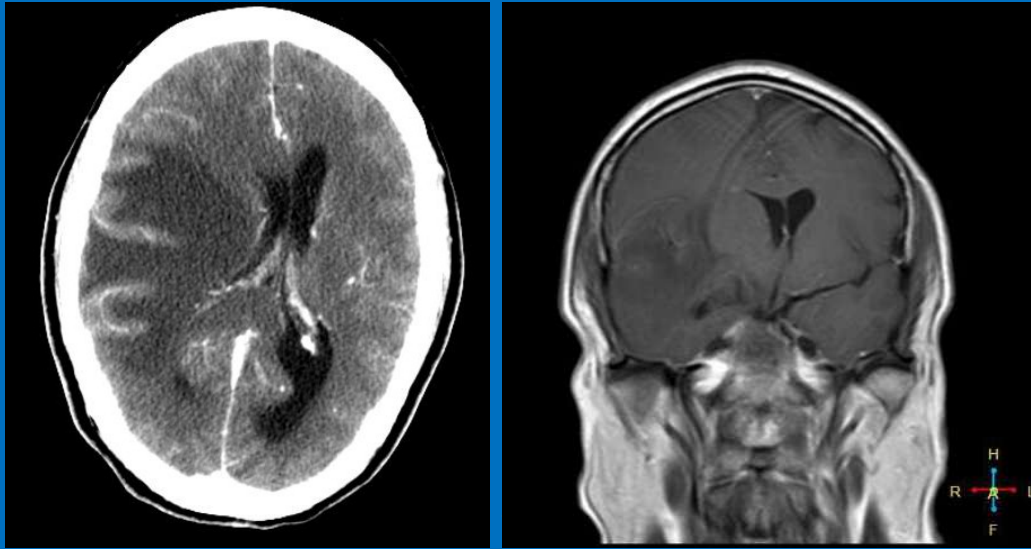


Herniation pathways: 1) subfalcine (cingulate gyrus), 2) uncal (transtentorial temporal), 3) Central (tentorial), 4) transcalvarial (external), 5) tonsillar (downward cerebellar)

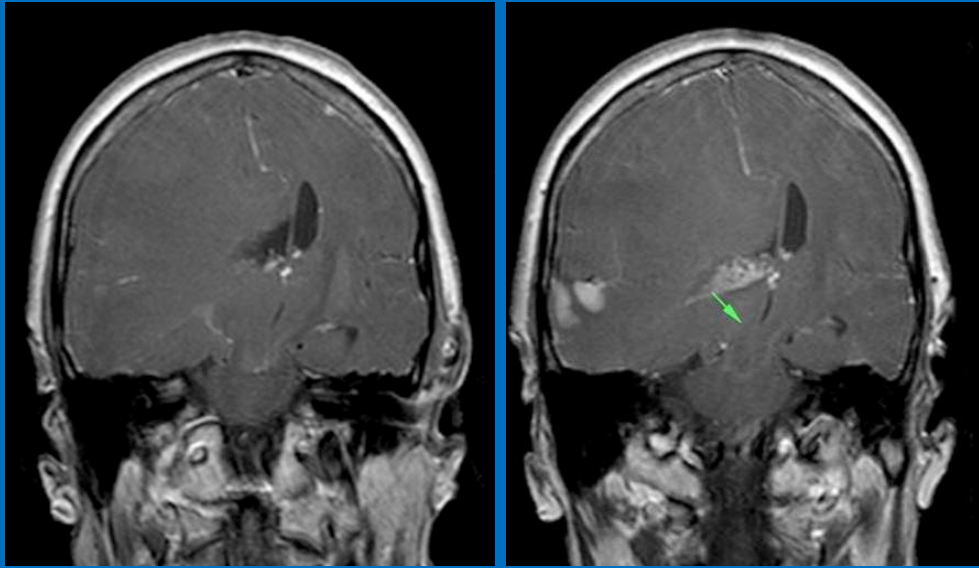
14

"Schematic" representation of various herniation pathways (1) cingulate gyrus, (2) uncal (temporal), (3) central, (4) cerebellar)

SUBFALCINE HERNIATION



TRANSTENTORIAL HERNIATION



DIFFERENTIAL OF IMAGING CHARACTERISTICS OF PRIMARY BRAIN TUMORS AND METASTASES

- METASTASES
 - Primary tumor and metastasis cannot be always differentiated
 - Multiple ring enhancing lesions are more likely to be metastatic
 - History of malignancy elsewhere makes metastasis more likely
- ABSCESS
 - Usually can not be differentiated by imaging
 - Clinical presentation of fever, elevated WBC
- STROKE/ HEMORRHAGE
 - May conform to a vascular distribution

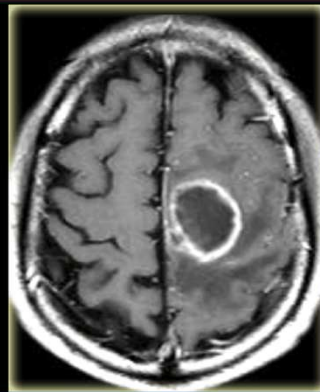
REMEMBER THIS?

**LEAKY VESSELS PRODUCE RING ENHANCEMENT ON
CONTRAST CT**

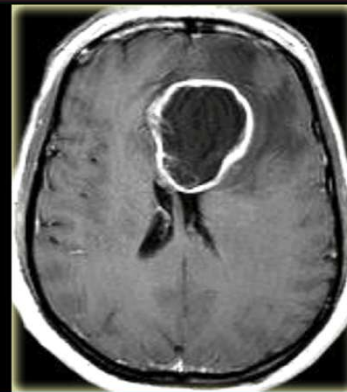
- YOU CANNOT ALWAYS TELL THE DIFFERENCE BETWEEN METASTASIS, ABSCESS, AND TUMOR IN THE BRAIN
- ALL HAVE BROKEN BLOOD-BRAIN BARRIER BECAUSE OF LEAKY ABNORMAL VESSELS



METASTASIS



ABSCSS



GBM

GBM = GLIOBLASTOMA MULTIFORME ¹⁸

COMMON TERMS

Glioma:

Refers to all glial tumors in general but is often also used to mean astrocytoma

Intra-axial:

Lesion is in brain or spinal cord tissue, e.g. astrocytoma and oligodendroglioma

Extra-axial:

Lesion is extrinsic to brain, e.g. meningioma and schwannoma

Anaplasia:

Cellular atypia and loss of differentiation associated with malignant tumors

CNS TUMORS IN A NUTSHELL

INCIDENCE:

- 2.4% (ca 13,000) of all cancer deaths are due to CNS tumors
Intracranial tumors: 10-17: 100,000; intraspinal tumors: 1-2: 100,000
- The incidence has a peak in childhood and then declines till 25 years; the CNS is among the three leading sites of cancer mortality in the first three decades of life
- In children brain tumors make about 20% of all cancers but are now the deadliest neoplasms accounting for about 26% of all cancer deaths in children
(The most common childhood cancers are leukemias; about 30% of all childhood cancers and cause about 25% of cancer deaths.)
- In adulthood there is a continuous increase with advancing age
- An apparent rise in the incidence in recent years has been attributed to more accurate diagnosis, especially with MRI imaging; some investigators argue that the rise is real
- Inarguably the incidence of cerebral lymphoma has increased

CNS TUMORS IN A NUTSHELL

INCIDENCE continued:

- The most common primary brain tumors are meningioma, astrocytoma, and glioblastoma multiforme (aka. anaplastic astrocytoma WHO grade IV)
- In children primary brain tumors are more common than metastatic tumors
- In adults metastatic brain tumors are more common than primary brain tumors
- The most common brain tumors in children are pilocytic astrocytoma and medulloblastoma
- Most brain tumors in children are infratentorial, located in the cerebellum and brainstem
- Most brain tumors in adults are supratentorial, located in the cerebrum
- The most common tumors of the spinal cord are schwannoma, meningioma, and ependymoma

CNS TUMORS IN A NUTSHELL

UNIQUE CHARACTERISTICS

Histology:

Morphologic differences between benign and malignant tumors are more subtle than in other organs.

Biologic behavior:

Well-differentiated low-grade lesions with low mitotic (= growth) rate and uniform cellularity may destroy important areas of brain and lead to significant morbidity and mortality and poor prognosis

Anatomic site:

Regardless of the type and differentiation of the tumor, in certain locations (e.g., medulla oblongata) it may cause death by compressing vital centers

Pattern of spread:

Extracranial spread of intracranial tumors is extremely rare
Metastatic spread to the CNS from extracranial tumors is common

GENETIC TUMOR SYNDROMES

- Account for a small proportion (1%- 4%) of CNS tumors

VON RECKLINGHOUSEN NEUROFIBROMATOSIS VRNF- Neurofibromatosis type 1-NF1

- Autosomal dominant inheritance pattern
- Caused by mutations of a gene on chromosome 17q that encodes **neurofibromin**
- Neurofibromin is involved in control of cell proliferation and **acts as a tumor suppressor of RAS oncoprotein**

BILATERAL ACOUSTIC NEUROFIBROMATOSIS - BANF - Neurofibromatosis type 2-NF2

- Autosomal dominant inheritance pattern
- Acoustic and spinal Schwannomas

GENETIC TUMOR SYNDROMES, continued

TUBEROUS SCLEROSIS

- Autosomal dominant inheritance pattern
- Clinical characteristics:
 - mental retardation, seizures
 - subependymal giant cell astrocytomas, cortical hamartomas (tubers) of the CNS
 - tumors and hamartomas of other organs
- Two mutations:
 - TS1 on 9q which encodes **hamartin**
 - TS2 on 16p which encodes **tuberin**
 - **Both acts as tumor suppressors**

GENETIC TUMOR SYNDROMES, continued

VON HIPPEL LINDAU DISEASE

- Autosomal dominant inheritance pattern
- Associated with:
 - hemangioblastomas of the cerebellum and retina
 - cysts of the liver and pancreas
 - Pheochromocytomas
 - kidney tumors
 - pancreatic tumors
- Linked to *VHL*, a tumor suppressor gene on chromosome 3p
- pVHL is involved in the rate of mRNA transcription
 - During transcription, cellular proteins can intervene in one of three stages: initiation, elongation, and termination.
- pVHL blocks the protein that stimulates elongation.
- pVHL inhibits vascular endothelial growth factor (VEGF)

IMPORTANT FACTORS IN THE PATHOGENESIS OF BRAIN TUMORS

AGE:

- Cerebellum is the most frequent site of CNS tumors children, cerebrum is the most frequent site of CNS tumors in adults.
- Embryonal tumors of the brain (cerebellar medulloblastoma) and other organs (adrenal neuroblastoma) occur predominantly in children.
- Explanation:
 - Neurogenesis and neuronal migration in the cerebrum are largely completed by midgestation, but in the cerebellum they continue for the first year of life.
 - Cerebellum is the most cellular part of the CNS (there are more granular neurons than all other neurons in the CNS together) and takes the longest to develop.
 - Glial cells are actively produced in cerebellum in childhood and the brisk cell division allows for new gene defects to arise and for inherited defects to be unmasked.

IMPORTANT FACTORS IN THE PATHOGENESIS OF BRAIN TUMORS continued

SEX:

- Males have more CNS tumors overall
- Females have more meningiomas

RADIATION:

- An increased incidence of CNS tumors, especially meningiomas, has been reported in patients who have received even low-dose radiation to the head.
- Children with ALL who have been treated with craniospinal radiation are at high risk for developing meningiomas and gliomas.
- The frequency of PNETs, including medulloblastoma, is increased in children, whose mothers had abdominal x-rays during pregnancy.
- Tumors emerge many years after radiation has been given and their incidence appears to be rising.

IMPORTANT FACTORS IN THE PATHOGENESIS OF BRAIN TUMORS *continued*

CHEMICAL CARCINOGENS:

- **Nitroso compounds (NOCs)**
 - The most potent neurocarcinogens in experimental animals
 - Present in foods (cured meats, fish, vegetables), cosmetics, rubber products, beer and water
 - Synthesized in the mouth, stomach, and bladder by nitrosation of dietary amines and amides, therefore they are ubiquitous
- **Formaldehyde**
 - Pathologist and funeral directors have more CNS cancers than the general population
- **Vinyl chloride**
 - Plastics manufacture
- **Acrylonitrile**
 - Textiles and plastics manufacture

IMPORTANT FACTORS IN THE PATHOGENESIS OF BRAIN TUMORS continued

IMMUNOSUPPRESSION:

Cerebral lymphomas, usually B-cell, are frequent in patients with:

- AIDS
- Renal transplants
- Congenital immunodeficiency syndromes
- Immunosuppressed states

EBV-INFECTION:

Some CNS lymphomas contain EBV-DNA

DIAGNOSIS OF CNS TUMORS

CLINICAL PRESENTATION:

- General symptoms and signs
- Thorough neurologic examination

IMAGING STUDIES:

- MRI
- CT

LABORATORY EVALUATION:

- Requires brain biopsy
- Pathological diagnosis is based on gross, microscopic (growth pattern, and vascularity, special features, such as rosettes), immunohistochemical, and ultrastructural features
- Assessment of histologic grade is based on:
 - Cellularity
 - Nuclear atypia
 - Proliferative index (mitotic count or Ki-67, which identifies proliferating cells, even if they are not in mitosis)
 - Necrosis
- Chromosomal and molecular analysis

MOLECULAR MARKERS IN CNS TUMORS

IDH (*IDH 1* and *IDH2* isocitrate dehydrogenase) gene mutations

- increased production of 2-hydroxyglutarate
- Present in low grade astrocytomas and oligodendrogliomas

Mutations in the telomerase promoter gene *hTERT*:

- In astrocytomas including glioblastomas
- Uncommon tumors with *IDH* mutations, which have LOF in *ATR*X (normally suppresses alternative lengthening of telomeres)

Co-deletion of 1p and 19q chromosomal segments:

- Oligodendrogliomas

Loss of 22q:

- meningioma
- Schwannoma (also mutations in the *NF2* gene)
- atypical teratoid-rhabdoid tumor

PTCH1 mutations :

- Important in cerebellar granular cell tumors (e.g., medulloblastoma)

31

PTCH1 is also implicated in nevoid basal cell carcinoma syndrome and other skin tumors, esophageal squamous cell carcinoma and transitional cell carcinomas of the urinary tract.

MOLECULAR MARKERS IN CNS TUMORS CONTINUED

- Genetic alterations commonly found in tumors outside the CNS are also found in CNS tumors
- The more malignant the tumor (e.g., GBM), the more mutations
 - Loss of heterozygosity (LOH) 10q (60-90%)
 - in GBMs but not in lower grades
 - *EGFR* amplification (40-50%)
 - more common in primary GBM
 - *MDM2* amplification/overexpression (10-15% of GBM)
 - *P53* mutations or deletion (20-40%)
 - *RB* mutations or deletion
 - *P16INK4a* mutation
 - *PTEN* mutation
 - *N-Myc* mutation
 - Up-regulation of *cyclin D2*

CLASSIFICATION OF BRAIN TUMORS

TUMORS OF NEUROGLIAL CELLS

A. TUMORS OF GLIAL CELLS

- ▣ Fibrillary astrocytoma
- ▣ Glioblastoma multiforme (anaplastic astrocytoma, grade IV)
- ▣ Midline glioma
- ▣ Oligodendroglioma
- ▣ Ependymoma - choroid plexus papilloma

B. NEURONAL TUMORS

- ▣ Ganglioglioma
- ▣ Gangliocytoma
- ▣ Central neurocytoma

C. EMBRYONAL TUMORS

- ▣ Medulloblastoma and other CNS PNETs

TUMORS OF CRANIAL AND SPINAL NERVES

- ▣ Schwannoma
- ▣ Neurofibroma

MENINGEAL TUMORS

- ▣ Meningioma

MESENCHYMAL TUMORS

- ▣ Sarcoma
- ▣ Hemangioblastoma

CEREBRAL LYMPHOMAS

GERM CELL TUMORS

- ▣ Teratoma
- ▣ Germinoma
- ▣ Craniopharyngioma

TUMORS OF THE PITUITARY GLAND

- ▣ Pituitary adenoma

METASTATIC TUMORS

WHO GRADING SYSTEM FOR ASTROCYTOMAS

Grade I - Pilocytic astrocytoma:

Benign cytological features

Grade II - Low-grade astrocytoma:

Moderate cellularity, no anaplasia or mitotic activity

Grade III - Anaplastic astrocytoma:

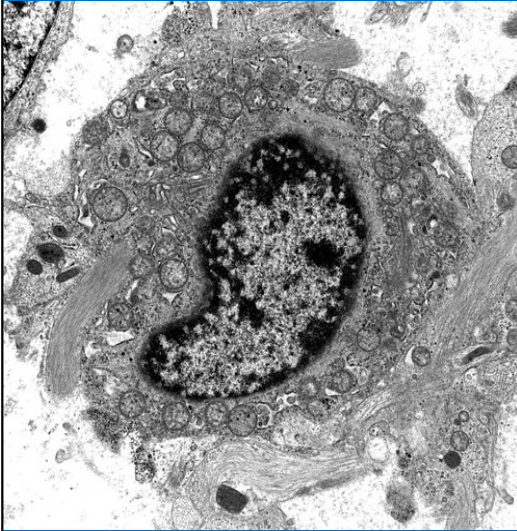
Increased cellularity, nuclear atypia, mitoses

Grade IV – Glioblastoma multiforme:

Same as Grade III plus vascular endothelial proliferation and necrosis

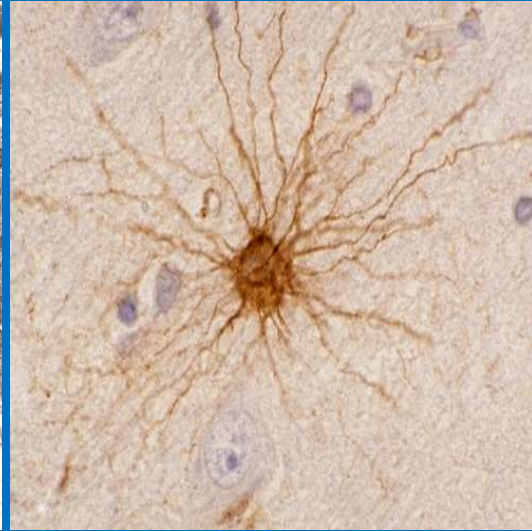
ASTROCYTOMA

INTERMEDIATE FILAMENTS



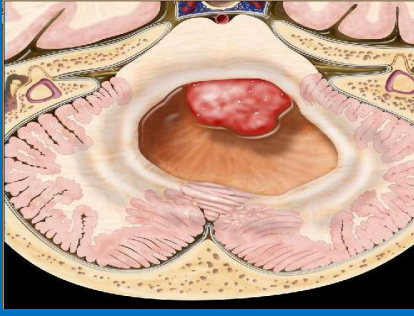
ASTROCYTE (TEM)

INTERMEDIATE FILAMENTS CONTAIN GLIAL
FIBRILLARY ACID PROTEIN (GFAP)

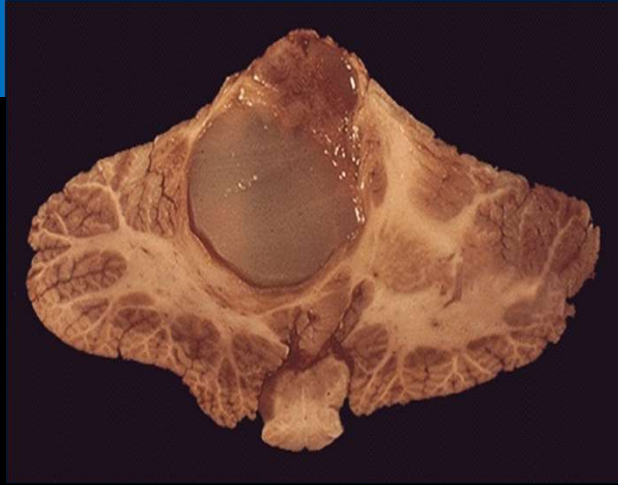
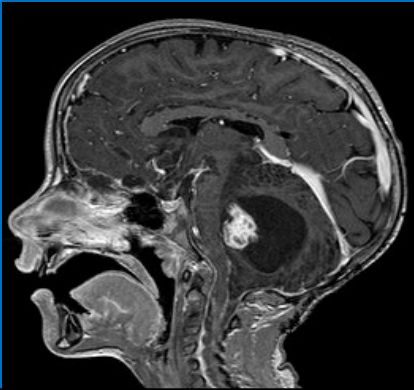


IMMUNOHISTOCHEMICAL STAIN FOR GFAP

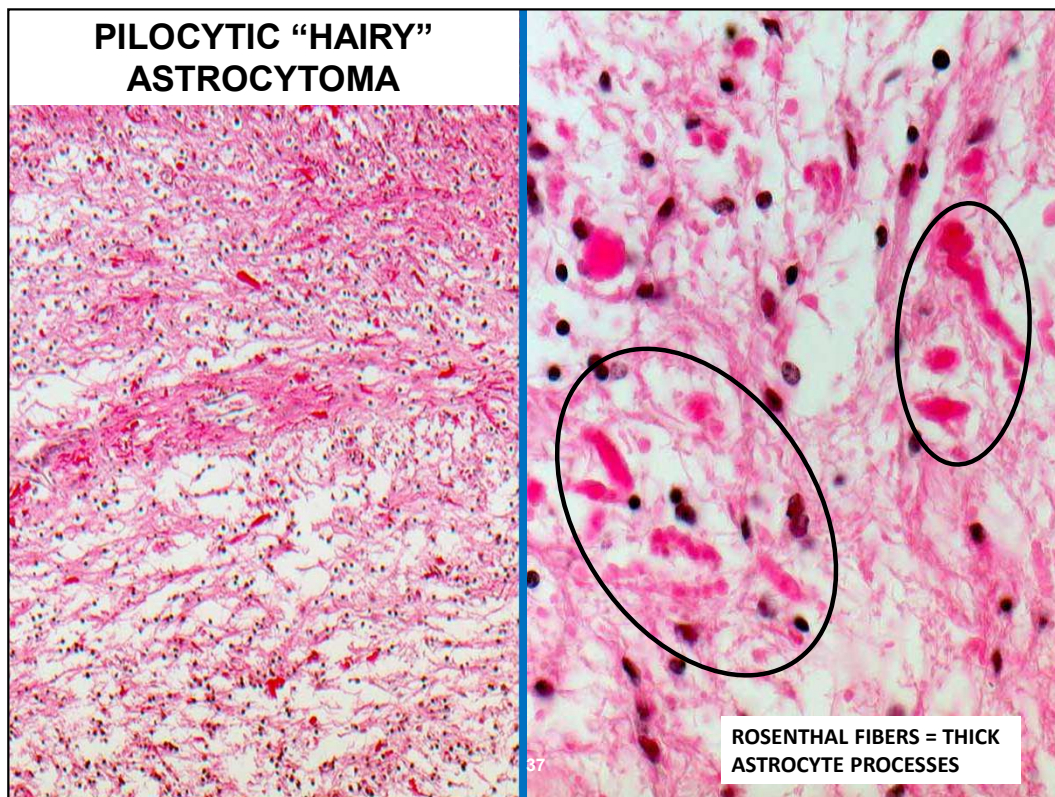
PILOCYTIC ASTROCYTOMA



- Lowest grade of astrocytoma
- Tumors of children and young adults
- Most arise in cerebellum and hypothalamus
- Some arise in the optic nerves and cerebral hemispheres and are more difficult to treat
- Often become cystic



Pilocytic astrocytoma

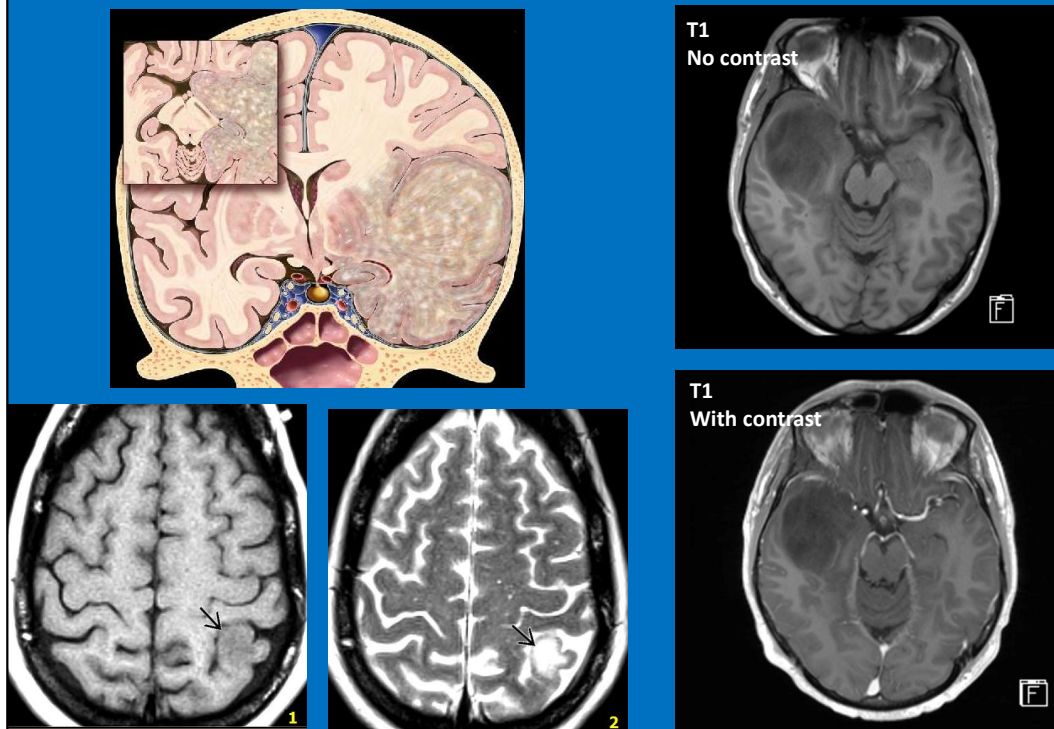


Clinical findings: 14 y o boy with a brain mass adjacent to 3rd ventricle.

Clinical diagnosis: Astrocytoma vs. craniopharyngioma. **Pathologic**

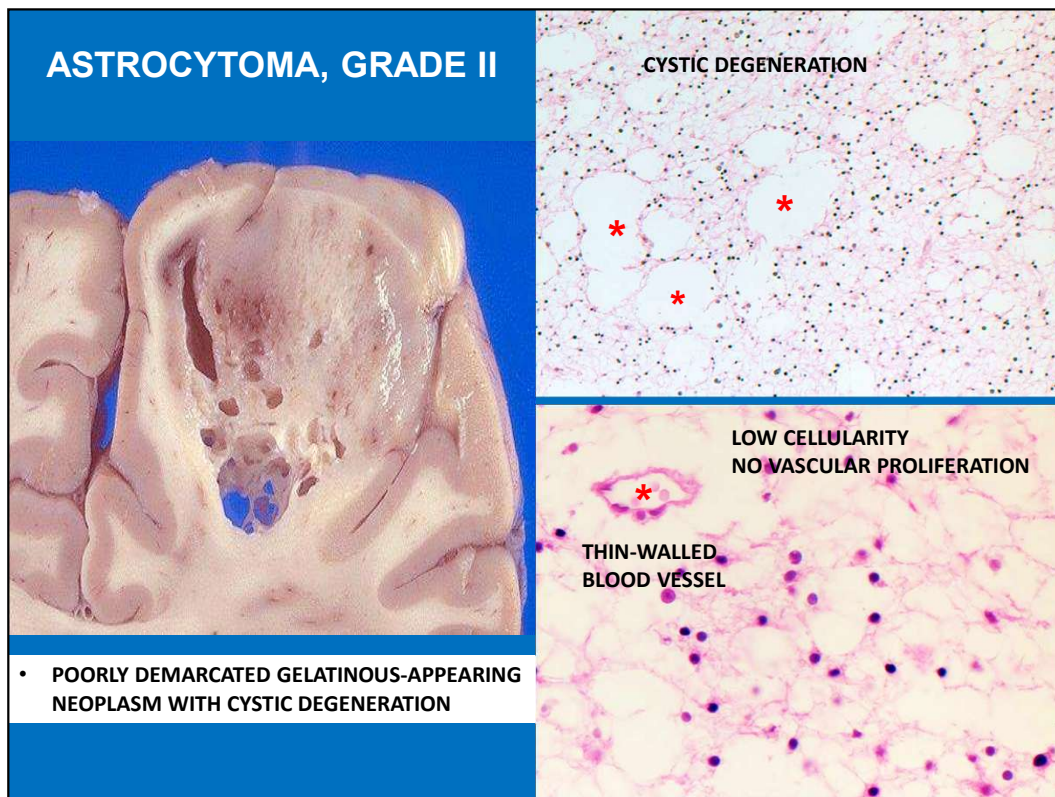
diagnosis: Pilocytic astrocytoma. **Description:** Parvicellular (cell poor) tumor with compact and loose microcystic areas and numerous Rosenthal fibers. Rosenthal fibers are particularly numerous in pilocytic (“hairy”) astrocytomas, but can be found also in benign reactive processes. **Stain:** HE.

FIBRILLARY ASTROCYTOMA, INFILTRATING, WHO GRADE II

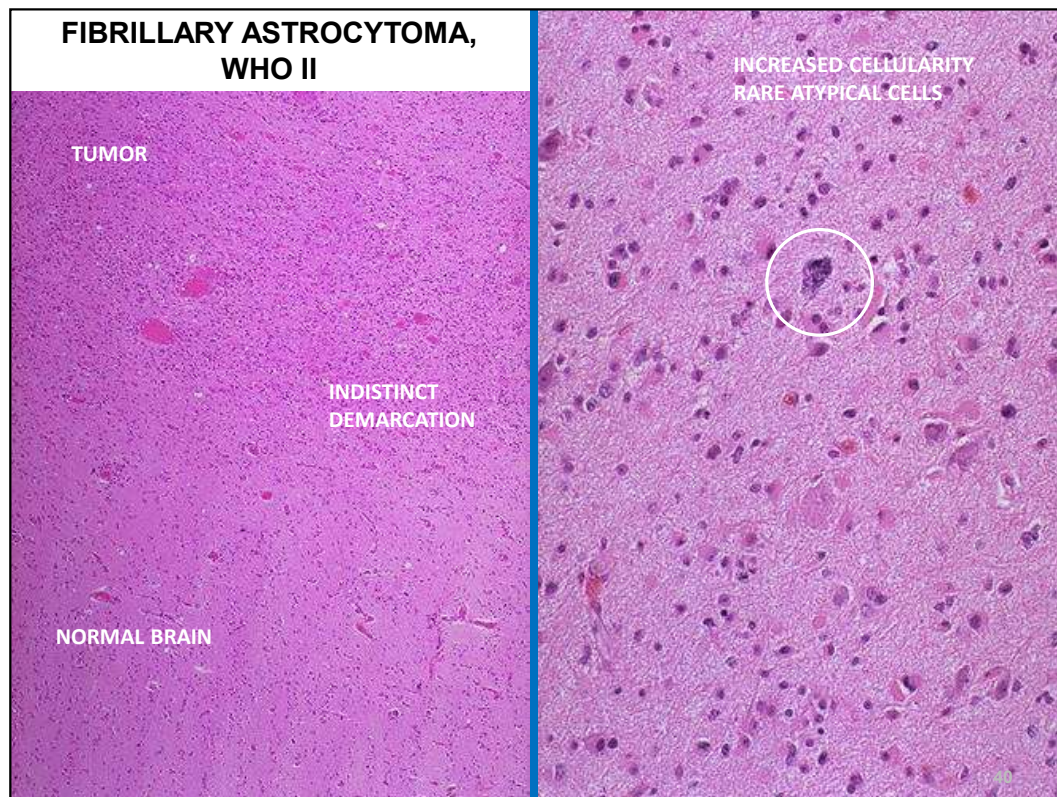


Left: Astrocytoma who grade 2. Left parietal lobe MRI T1 and T2 images. Astrocytomas can occur anywhere but are most frequent in the cerebral hemispheres. Most are poorly demarcated. Borders are difficult to determine during surgery. Difficult to differentiate histologically from reactive gliosis in frozen sections.

Right: Fibrillary Astrocytoma, diffuse mass with not significant enhancement after IV Gadolinium is given.

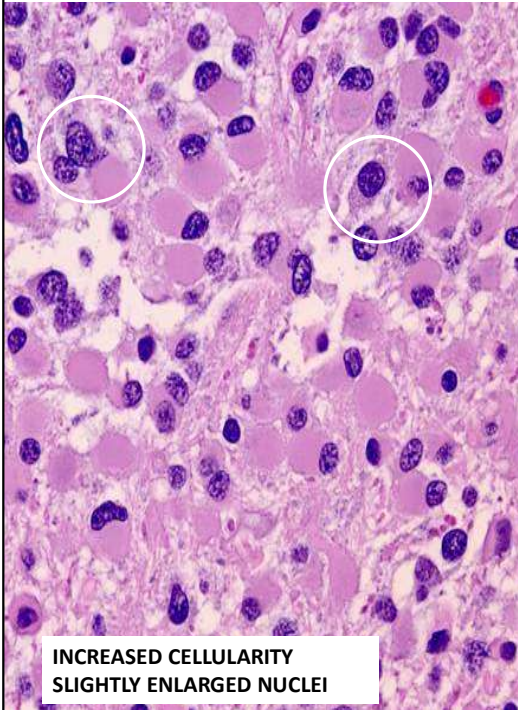


Left: Brain, gross. Astrocytoma. Astrocytomas often undergo cystic degeneration. **Right:** Fibrillary astrocytoma WHO II. **Description:** Microcystic, tumor with low cellularity in a fibrillary matrix. No vascular proliferation or necrosis. This is a typical low-grade astrocytoma histologically. **Stain:** HE.

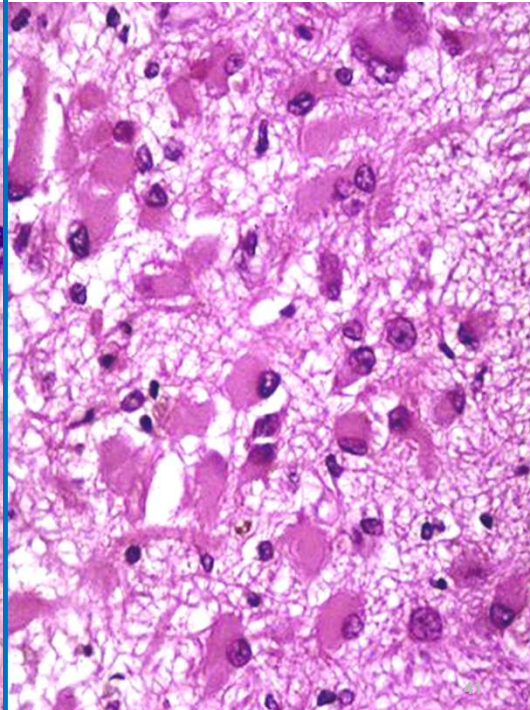


Low grade infiltrating astrocytoma (glioma) on top, normal brain on bottom.
Note increased cellularity and indistinct margin of the tumor.

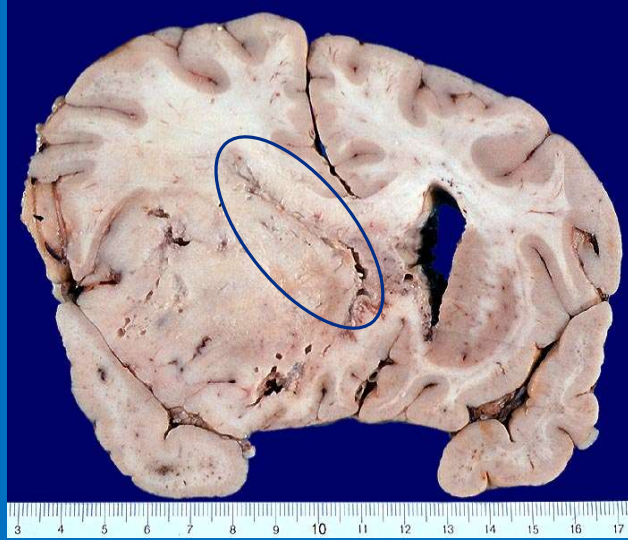
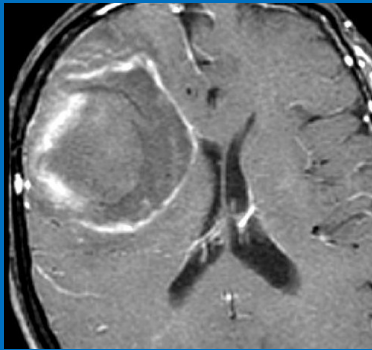
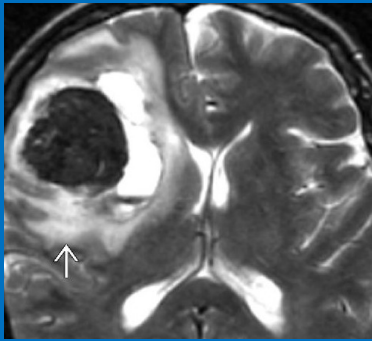
**VARIANT: GEMISTOCYTIC
ASTROCYTOMA, WHO II**



**COMPARE: REACTIVE
GEMISTOCYTIC ASTROCYTES**



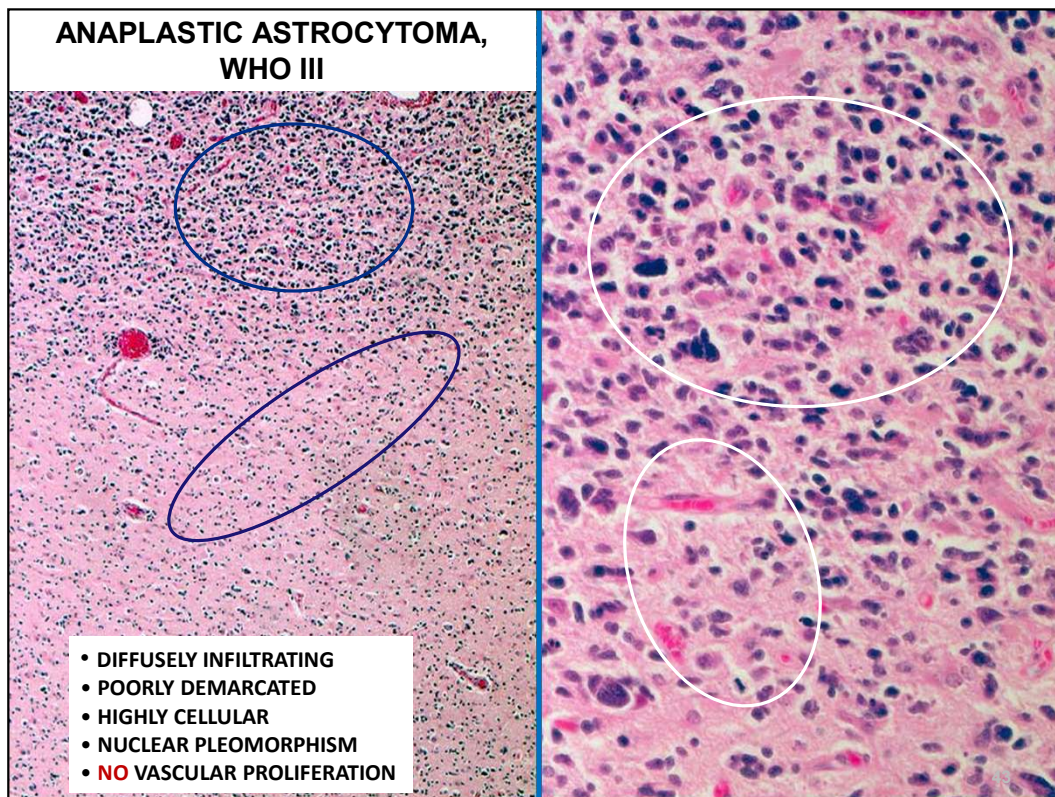
ANAPLASTIC ASTROCYTOMA, WHO GRADE III



- POORLY DEMARCATED GRAY-WHITE TUMOR IN LEFT FRONTOTEMPORAL AREA.
- THE HORN OF THE LATERAL VENTRICLE IS COMPLETELY OBLITERATED BY THE TUMOR.

42

Left: Grade 3 Astrocytoma. T2 and T1 post contrast enhancement images. Hemorrhage in the T2 image is causing signal void (black area arrow)



Clinical findings: 44 y o woman with a recurring brain tumor. **Pathologic diagnosis:** Anaplastic astrocytoma, WHO grade III. **Description:** Diffusely infiltrating poorly demarcated highly cellular tumor composed of pleomorphic cells with atypical nuclei. Stain: HE.

GLIOBLASTOMA MULTIFORME (GBM) ANAPLASTIC ASTROCYTOMA, WHO IV

- The most malignant glioma.
-
- Most common in middle aged or older adults but may occur at any age.
- Most common sites are the frontal and temporal lobes, but GBM may involve any part of the CNS.
- Most GBMs arise de novo (primary GBM), some arise by malignant transformation from low-grade astrocytomas (secondary GBM).
- Pseudopalisading of tumor cells around central necrosis and vascular proliferation are useful morphologic diagnostic features.
- Primary GBMs are often composed of small undifferentiated cells (small cell GBM), show extensive ischemic necrosis and have a higher proliferative index.
- Primary GBMs are more common in older patients and are more aggressive.

GLIOBLASTOMA MULTIFORME (GBM) ANAPLASTIC ASTROCYTOMA, WHO IV

Continued

- Secondary GBMs are composed of larger cells with astrocytic differentiation.
- For adults with more aggressive glioblastoma median survival is about 14.6 months and two-year survival is 30%; almost 10% of patients with GBM may live five years or longer
- Glioblastomas are generically heterogeneous, therefore outcomes in similar appearing cases may differ; e.g., glioblastoma patients, whose MGMT gene is shut off by methylation, have better survival rates.
- The MGMT gene is thought to be a significant predictor of response.
- Children with high-grade tumors (grades III and IV) tend to do better than adults; five-year survival for children is about 25%.

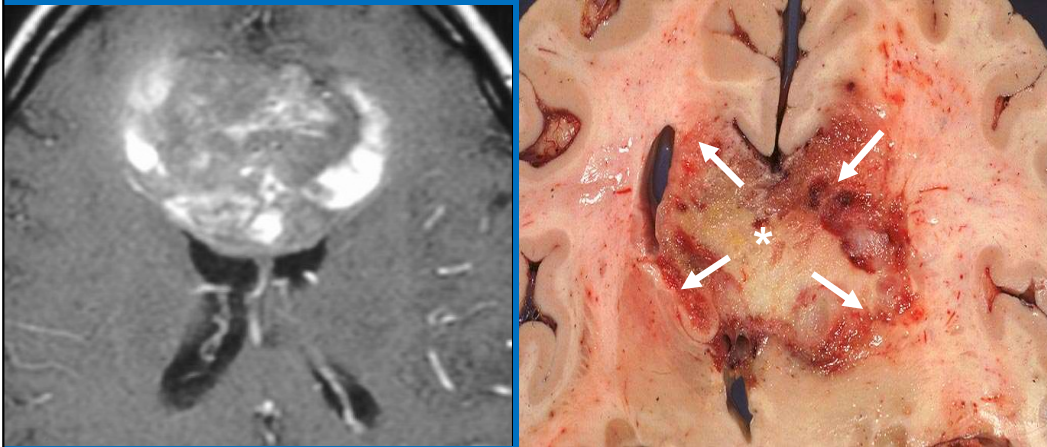
ANGIOGENESIS IN GBM

- Hypoxia develops as GBM outgrows its vascular supply.
- Hypoxia induces upregulation of hypoxia inducible factor 1 (HIF-1).
- HIF-1 in stimulates the expression of vascular endothelial growth factor (VEGF)
- VEGF stimulates formation of new vessels, which allows continuing tumor growth.
- The new vessels are often arranged in **glomeruloid formations** and lack a blood- brain barrier, which contributes to cerebral edema, a clinically important feature of GBM.

MOLECULAR EVENTS IN THE DEVELOPMENT OF GBM

- Mutation of the tumor suppressor gene *p53* on 17p is a pivotal event in the transformation of normal to neoplastic astrocytes.
- Secondary GBMs are characterized by early *p53* mutation and subsequent loss of the long arm of chromosome 10.
- Deletion of the tumor suppressor *PTEN* (Phosphatase and Tensin Homologue) in chromosome 10 with losses of several other tumor suppressor genes in other chromosomal loci convert low-grade astrocytomas to anaplastic astrocytomas and GBMs.
- Primary GBMs overexpress epidermal growth factor receptor (EGFR), a tyrosine kinase coded by a gene on 7p.
- Expression of *EGFR* and related genes determines the response of GBM to tyrosine kinase inhibitors, which are used in GBM chemotherapy.

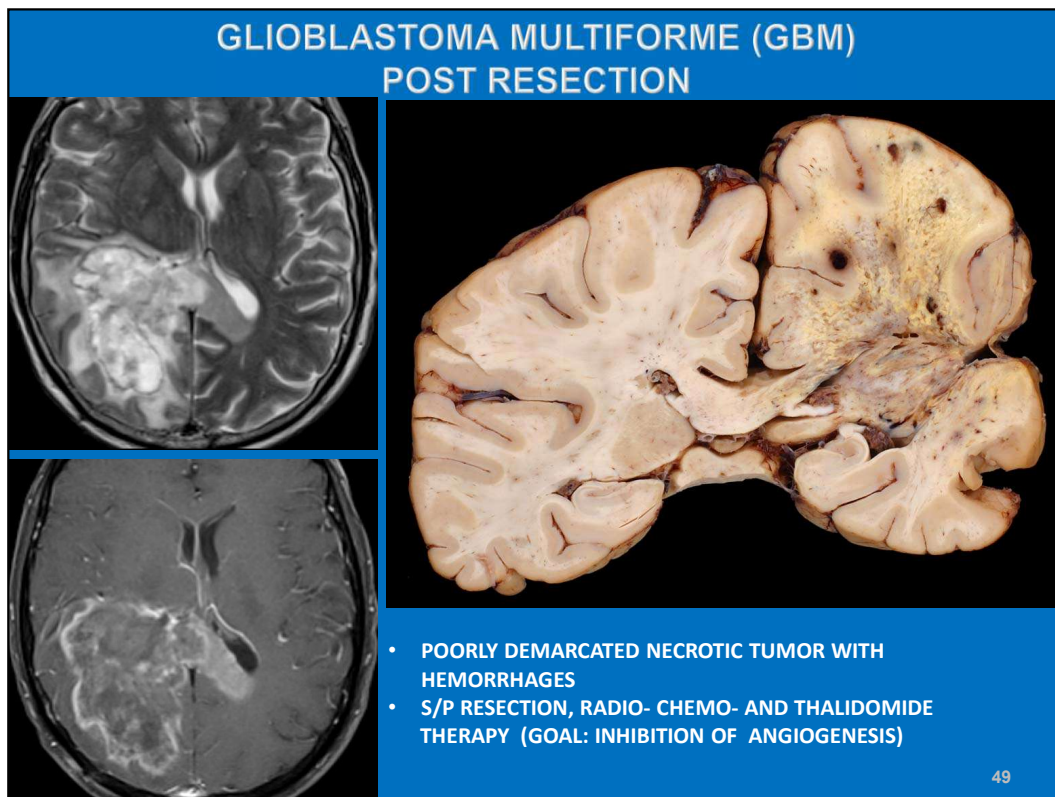
GLIOBLASTOMA MULTIFORME (GBM), ASTROCYTOMA, WHO GRADE IV



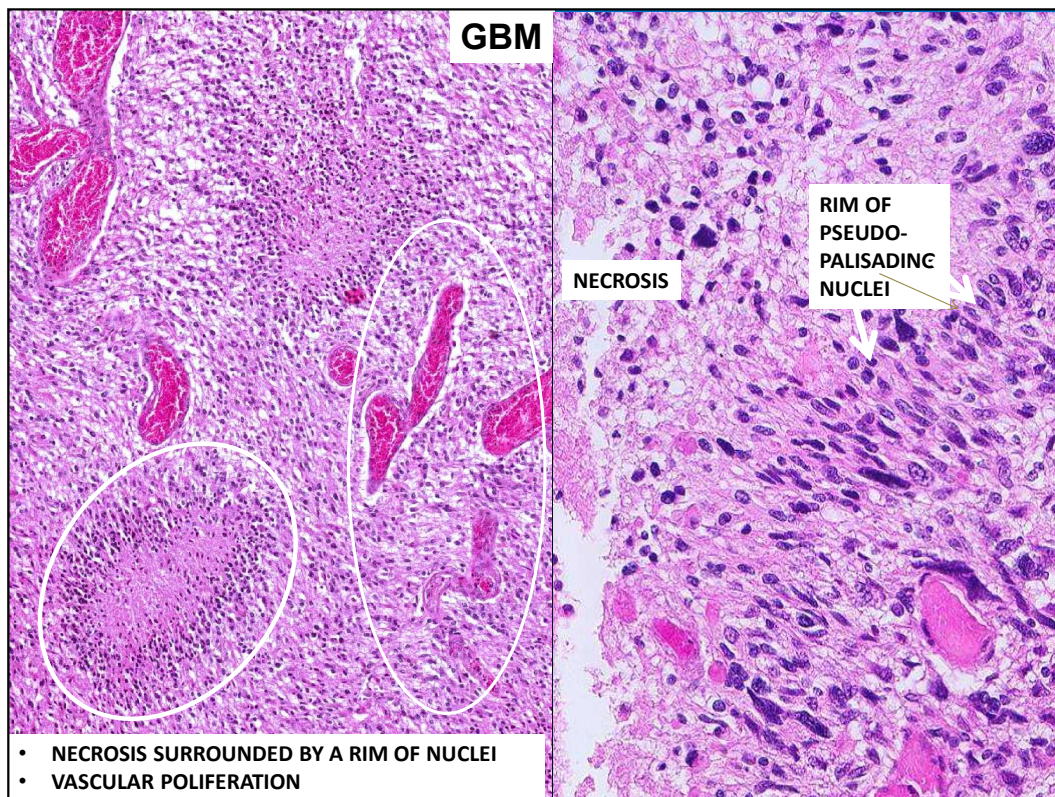
- POORLY DEFINED HEMORRHAGIC NECROTIC TUMOR
- DESTROYS CORPUS CALLOSUM
- INFILTRATES INTO THE HEMISPHERES
- COMPLETELY OBLITERATES THE LEFT LATERAL VENTRICLE.

48

Clinical findings: Neurologic symptoms for six months with aphasia, psychomotor slowing, and urinary incontinence. Imaging studies revealed a mass destroying corpus callosum. One month later the patient's status grew significantly worse with general symptoms of fever and malaise. Immediate cause of death was aspiration pneumonia.

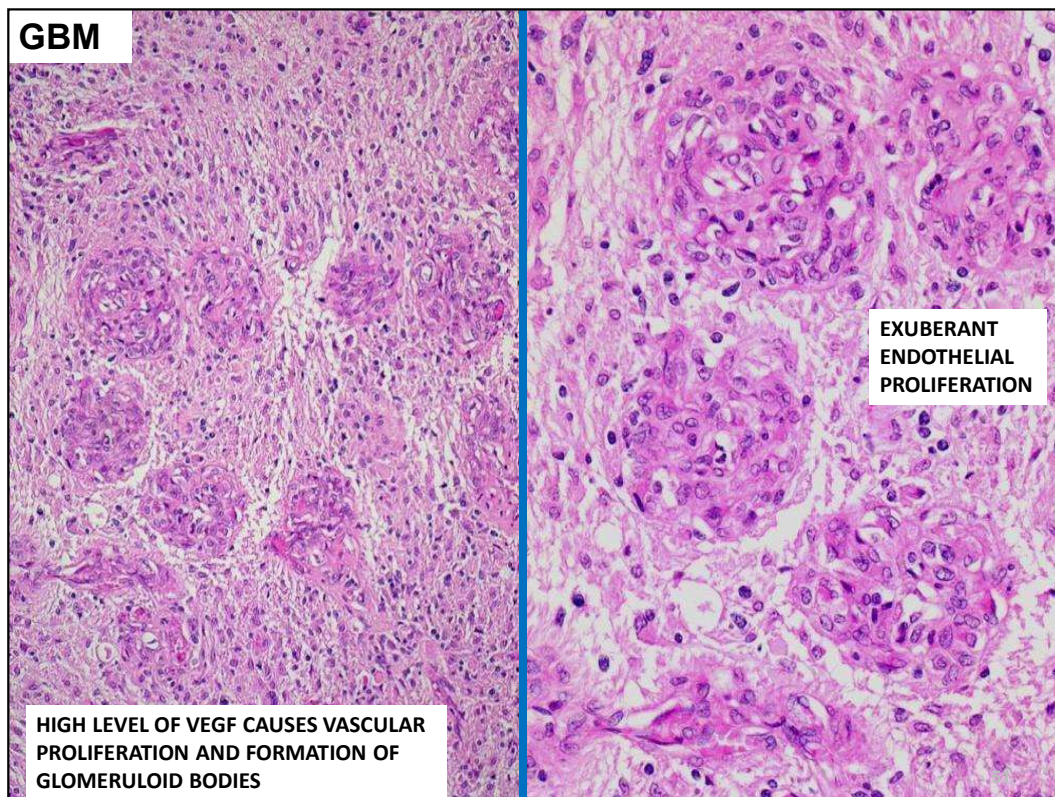


Left: GBM images MRI T2 and T1 with contrast enhancement. Poorly defined irregular margin with enhancement, surrounding edema, mass effect.



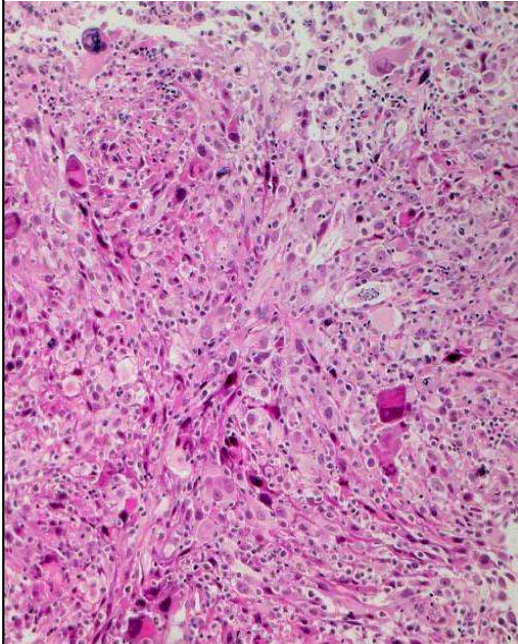
Left: 1340.svs. 10X. Highly cellular malignant tumor with prominent vascular proliferation and necroses surrounded by pseudopalisading tumor cells.

Right: 52 yow with a two month history of profound personality changes, headache, and progressive focal neurologic symptoms. **Diagnosis:** Glioblastoma (WHO IV). **Description:** Highly cellular (blue = lots of nuclei) tumor with extensive necrosis and an infiltrative growth pattern. **Stain:** HE.

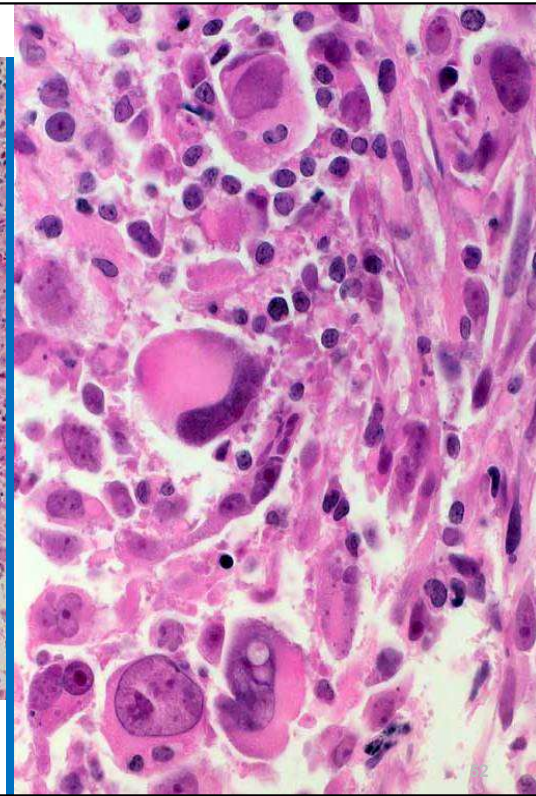


52 y o woman with a two month history of profound personality changes, headache, and progressive focal neurologic symptoms. **Diagnosis:** Glioblastoma multiforme (WHO IV). **Description:** Glomeruloid vascular proliferation.

VARIANT: GIANT CELL GBM



- 5 % OF ALL GBMs
- SLIGHTLY BETTER PROGNOSIS IN SPITE OF EXTREMELY MALIGNANT APPEARANCE



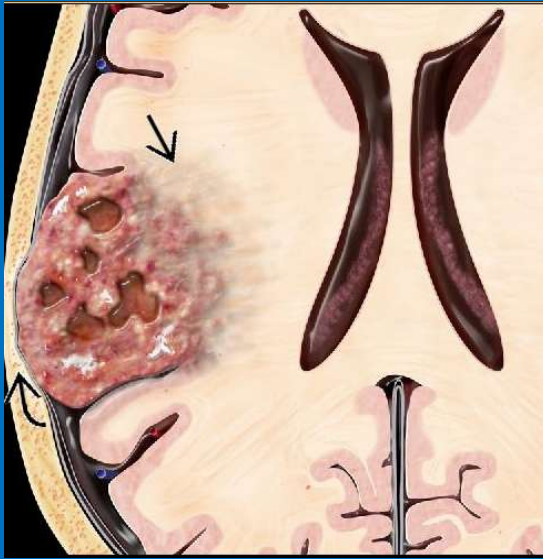
70 y o woman. Diagnosis: Giant cell glioblastoma multiforme (astrocytoma WHO IV). **Description:** The tumor is highly cellular and is composed of extremely pleomorphic population of cells. Some tumor cells are spindle-shaped and many have bizarre multilobed giant nuclei. **Clinical findings:** CT: Right parietal area glioma? **Comment:** Giant cell glioblastomas make up about 5% of all glioblastomas. In spite of the extremely malignant appearing histologic features, the prognosis is slightly better than in regular glioblastoma multiforme. **Stain:** HE.

ASTROCYTOMAS recap

- Account for 80% of adult primary brain tumors
- Most frequent in the 4th- 6th decades
- Usually located in the cerebral hemispheres in adults, in hypothalamus or cerebellum in children
- Clinical symptoms include headache and seizures; focal symptoms are determined by the location
- Histologic differentiation correlates with clinical course and outcome
- Well-differentiated astrocytomas grow slowly with a mean survival >5 years
- All astrocytomas have malignant potential and often develop more malignant features later during the course.
- Many astrocytomas present from the start as anaplastic astrocytoma, grade IV, aka. glioblastoma multiforme (GBM).
- GBMs have a median survival rate of 14.6 months with 30% 2-year survival rate.

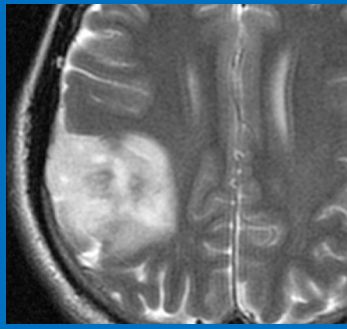
53

OLIGODENDROGLIOMA

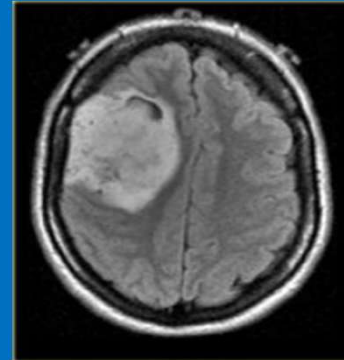


- 3-15% of gliomas
- Usual location in cerebral hemispheres
- Most common in the 4th and 5th decade
- Insidious, slow growing
- Mean survival 5 years
- Appears more circumscribed than astrocytomas, but infiltrates the cortex diffusely
- Often develops areas of cystic degeneration
- Often bleeds, because of rich capillary network
- Calcifications are common in the periphery of the tumor and help in preoperative radiologic diagnosis

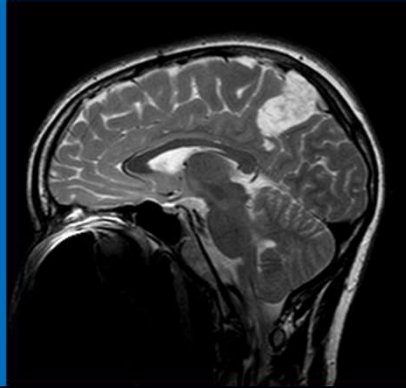
OLIGODENDROGLIOMA



MRI



CT



TUMOR

- Unilateral calcification

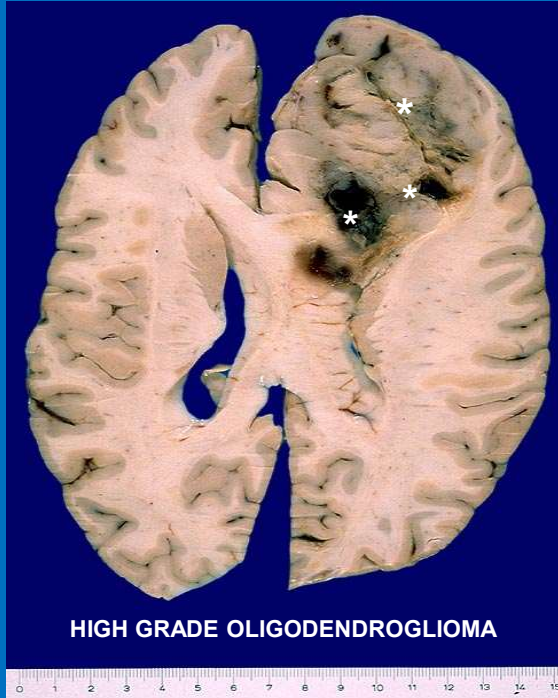
BILATERAL SYMMETRIC CALCIFICATIONS:

- Choroid plexus may be calcified in older individuals



Calcifications are seen on CT and may not be seen on MRI.

OLIGODENDROGLIOMA



HISTOLOGY:

- Uniform tumor cells with round nuclei surrounded by a halo (artefact of processing).
- Rich delicate capillary network predisposes to bleeding and later **calcification**, which is helpful in radiological and histological diagnosis.
- Classification as low-grade or high grade is based on cellularity, mitotic activity, vascular endothelial proliferation, and necrosis.

EM:

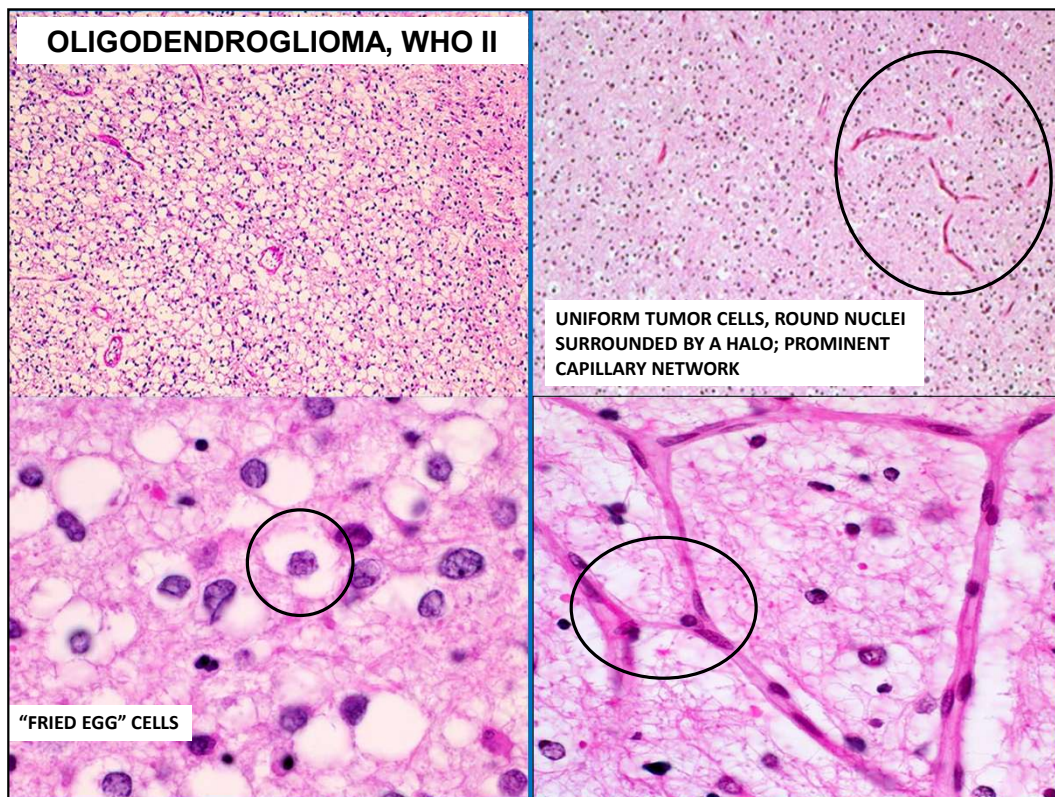
- **Abundant plasma membranes form concentric layers mimicking myelin.**

BIOLOGIC BEHAVIOR:

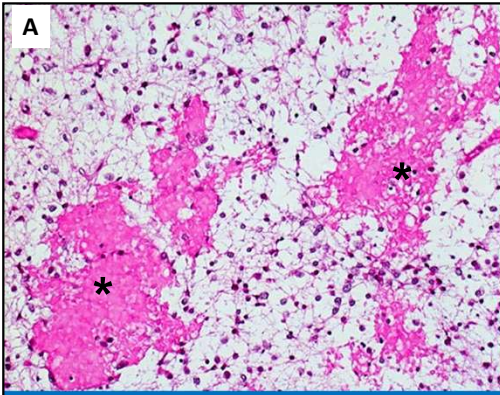
- Histologic parameters correlate with length of survival.
- Oligodendrogliomas are among the most **chemosensitive** solid tumors.
- Signs of high grade: diffuse infiltration of cortex, necrosis and hemorrhage.

MOLECULAR EVENTS:

- Losses of chromosomes 1p and 19q correlate with increased sensitivity to chemotherapy and longer survival.

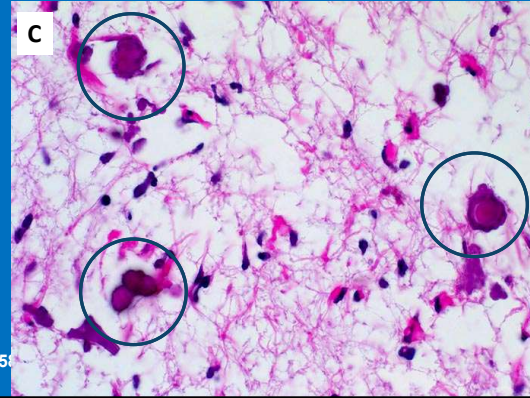
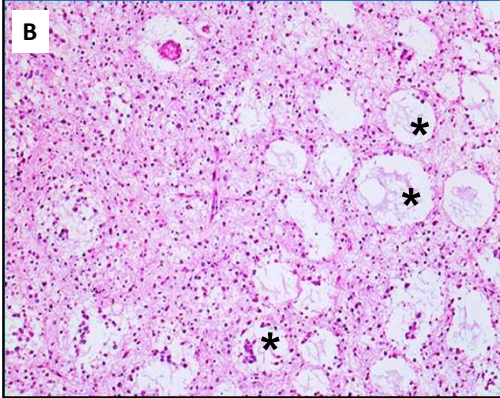


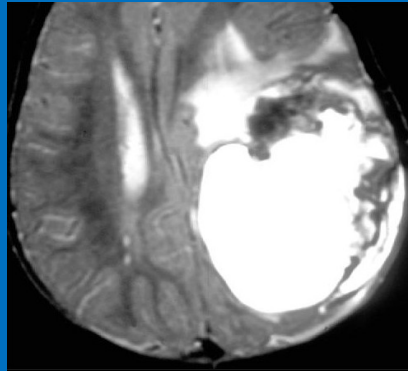
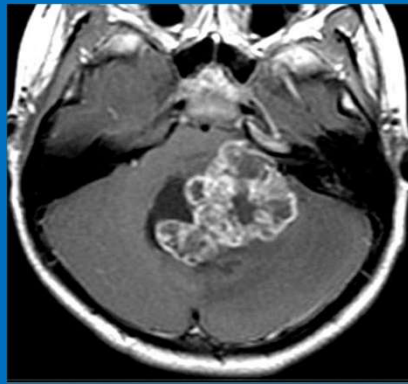
Diagnosis: Oligodendroglioma (WHO grade II). **Left:** Small tumor cell nuclei are surrounded by a halo (fixation artefact) and resemble slightly enlarged oligodendrocytes of the white matter. **Right:** The tumor contains a prominent branching capillary network. **Clinical findings:** Headache and epileptic seizures for two years. Stain: HE.



OLIGODENDROGLIOMA, WHO II

- A. Fragile capillaries bleed easily
- B. Cystic degeneration in the areas of prior bleed
- C. Dystrophic calcifications at the periphery of the tumor





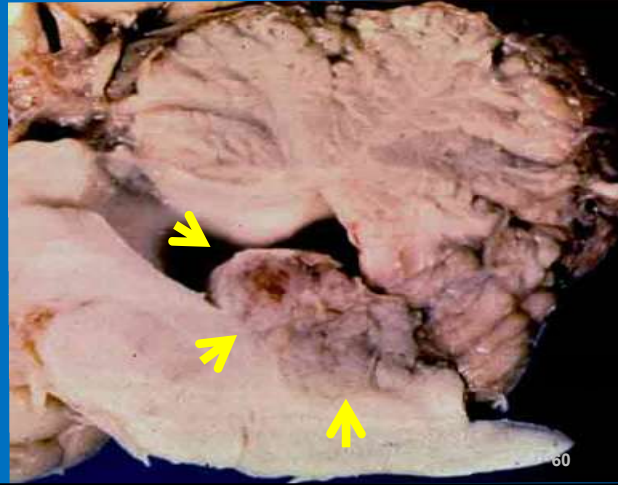
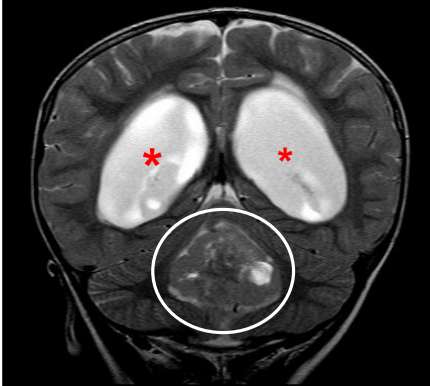
EPENDYMOMA

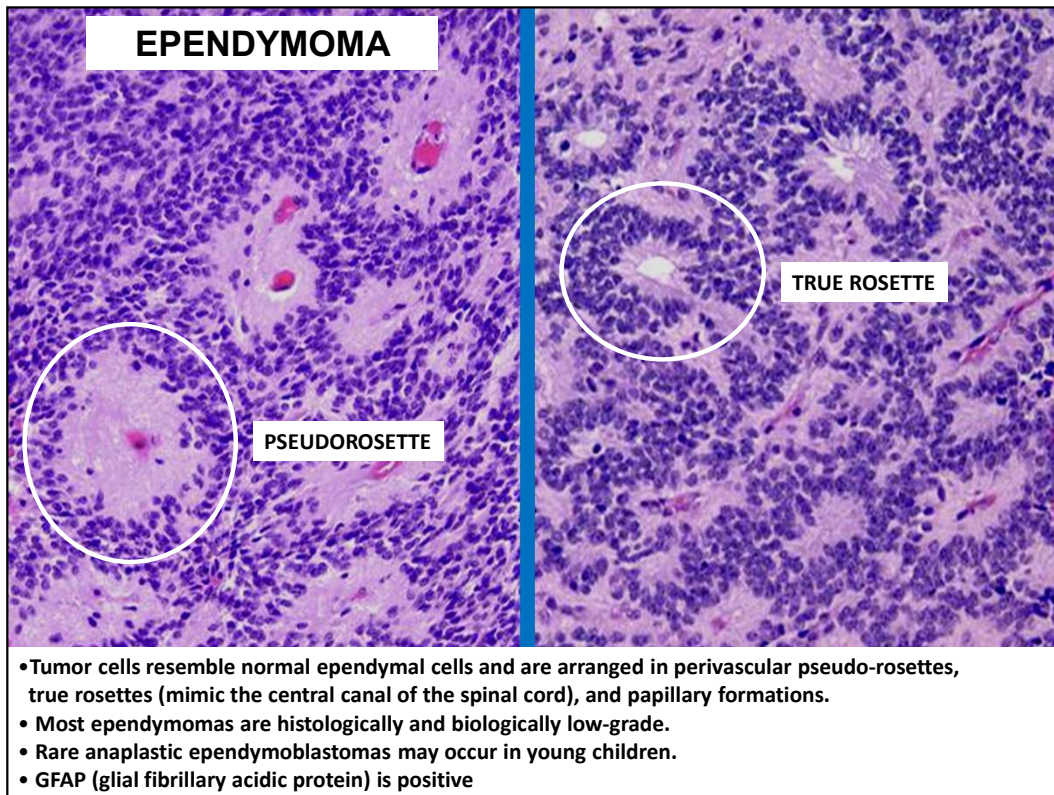
CLINICAL FEATURES:

- Tumor of children and young adults.
- Arises most frequently in the fourth ventricle
- Difficult to operate.
- Causes hydrocephalus by blocking CSF flow.
- Symptoms of an ependymoma are related to the location and size of the tumor.
- In neonates and infants, enlargement of the head may be one of the first symptoms.
- Irritability, sleepiness and vomiting may develop as the tumor grows.
- In older children and adults, nausea, vomiting and headache are the most common symptoms
- May occur anywhere in relation to the ventricular system or central canal.
- The most common primary intra-axial tumor in the spinal cord and filum terminale.

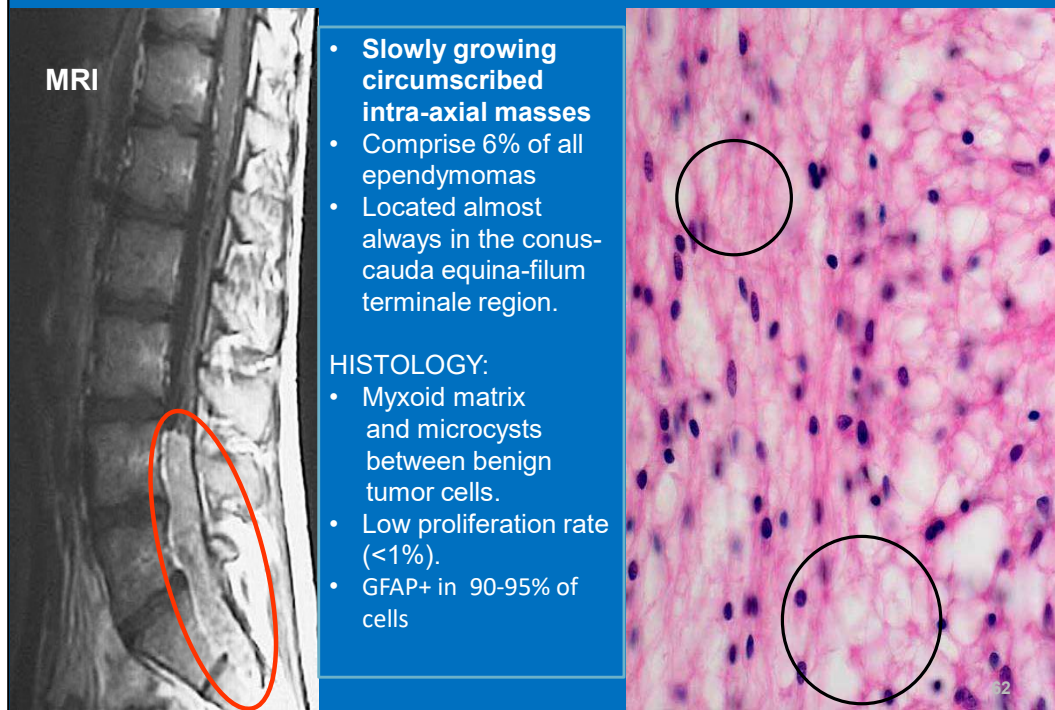
EPENDYMOMA

- **Tumor arises on the floor of the 4th ventricle**
- Exophytic, protrudes into and out of the 4th ventricle
- Well demarcated from the surrounding brain and spinal cord
- Blocks flow of CSF and causes hydrocephalus

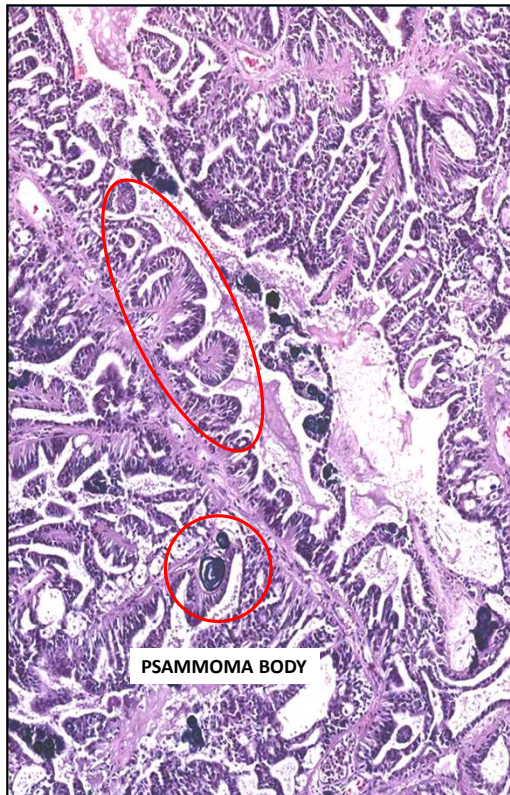




SPINAL MYXOPAPILLARY EPENDYMOMA



Diagnosis: Myxopapillary ependymoma. **Description:** Myxoid matrix and microcysts are seen between tumor cells. **Additional findings:** Ca. 90-95% of tumor cells stain for GFAP. The proliferation rate of this tumor is extremely low, far below 1%. **Clinical findings:** Spinal cord ependymoma in filum terminale. **Comment:** Myxopapillary ependymomas make up 6% of all ependymomas and they are practically always found in the conus-cauda equina-filum terminale region. **Stain:** HE.



CHOROID PLEXUS PAPILLOMA

- Most common in children
- Occurs anywhere along the choroid plexus
- In children in the lateral ventricles
- In adults in the fourth ventricle
- Often has psammoma bodies (lamellar calcifications)
- Papillary growths recapitulate the structure of the normal choroid plexus almost exactly: fibrovascular stalks are covered by cuboidal or columnar epithelium
- Presents with hydrocephalus due to obstruction of the ventricular system by tumor or overproduction of CSF

CHOROID PLEXUS CARCINOMA

- Resembles adenocarcinoma
- Rare
- More common in children than adults
- Important to differentiate from metastatic carcinoma in adults

63

PRIMITIVE NEUROECTODERMAL TUMORS (PNETs)

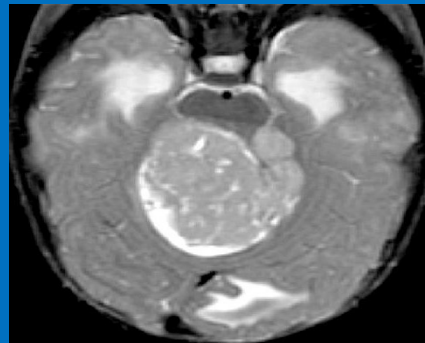
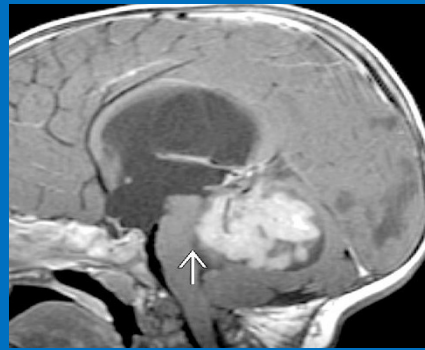
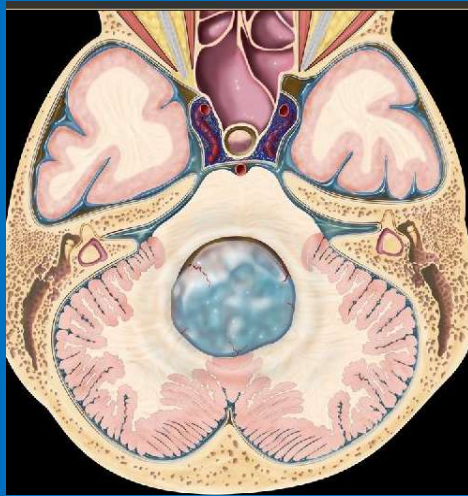
- This class of tumors contains Medulloblastoma and other "small blue round cell tumors" of the brain

EXAMPLE: MEDULLOBLASTOMA

- High incidence in infants and children
- The second most frequent CNS tumor in children after pilocytic astrocytoma
- Occurs in the first decade of life with a second peak in the early 20s
- Embryonal tumor, analogous to Wilms tumor and adrenal neuroblastoma ("small blue round cell tumor")
- Undifferentiated, immature appearance; resembles developing neural tissue
- Highly malignant
- Cells of origin: primitive cells of the external granular layer of the cerebellum, which persists until the beginning of the second year of life
- External granular layer of the cerebellum is composed of precursor cells that migrate from the roof of the fourth ventricle to the surface of the developing cerebellum where the cells divide and differentiate
- Neurons move inwards and form the permanent granular layer of the cerebellar cortex
- Remnants of the provisional granular layer are seen in the posterior and anterior medullary velum.

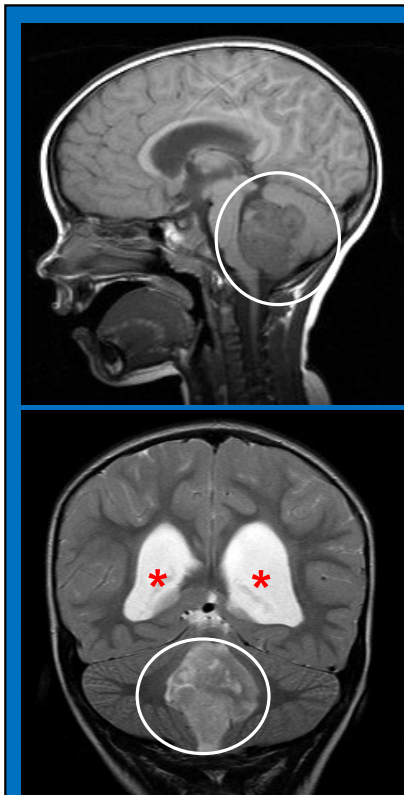
64

PRIMITIVE NEUROECTODERMAL TUMORS (PNETs)



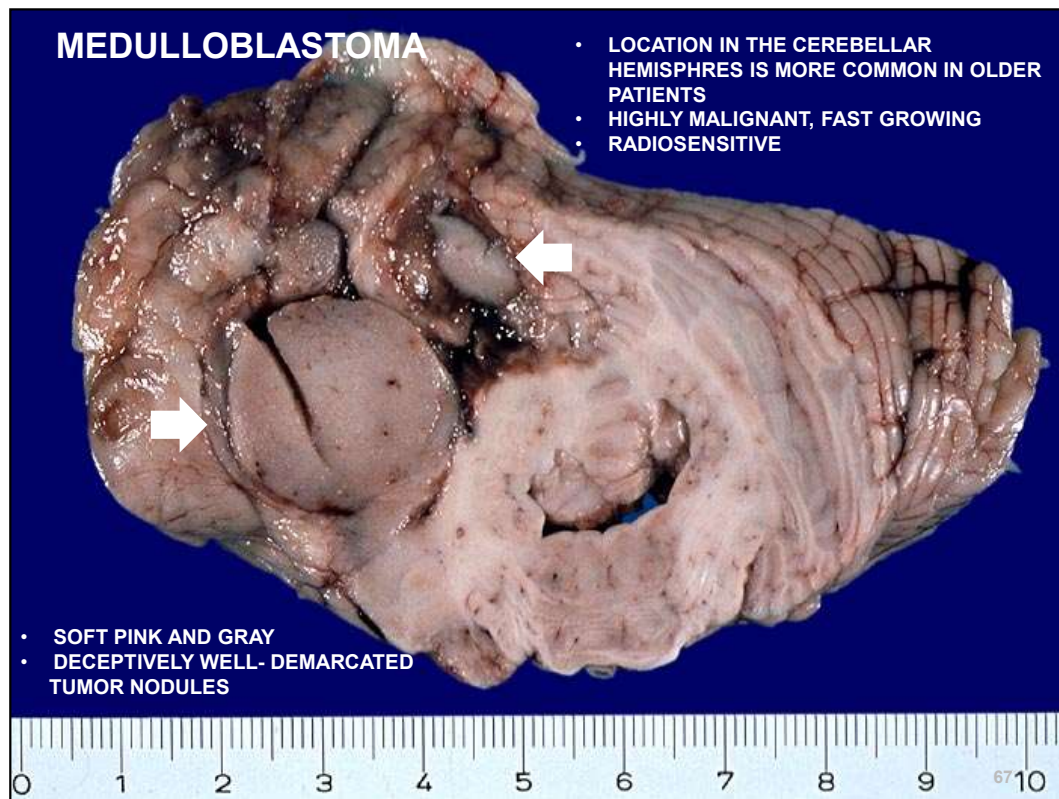
MRI:

- Compact
- Isointense
- Shows contrast enhancement



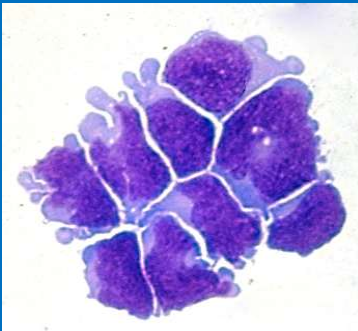
MEDULLOBLASTOMA

- Commonly presents with morning headache, vomiting and blurred vision due to increased intracranial pressure
- Funduscopy: papilledema
- Other symptoms: ataxia, strabismus, nystagmus, stiff neck (sign of tumor extension through the foramen magnum)
- Differential diagnosis: gastroenteritis, viral meningitis
- Lumbar puncture in this setting is contraindicated: can induce cerebellar tonsillar herniation
- Typical site: posterior vermis adjacent to the roof of the 4th ventricle
- In children arises usually in the midline
- Grows into and compresses the 4th ventricle
- Causes hydrocephalus



Diagnosis: medulloblastoma. **Description:** Gray, deceptively well demarcated tumor nodules in the left cerebellar hemisphere.

MEDULLOBLASTOMA

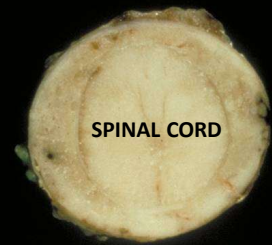
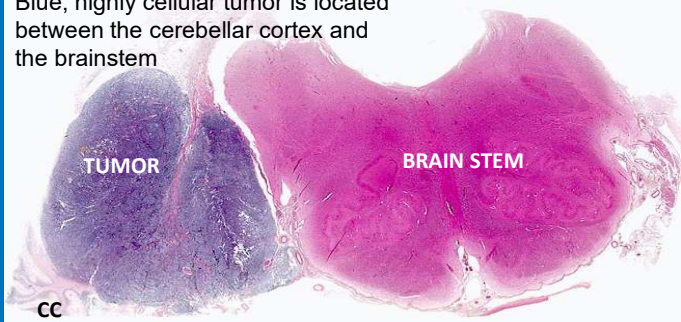


CSF:

High protein, low glucose, occasional lymphocytes and small tumor cells with scanty cytoplasm and round to oval nuclei

CSF cytology is used to monitor tumor, which tends to spread along the ventricular walls and to seed the subarachnoid space

Blue, highly cellular tumor is located between the cerebellar cortex and the brainstem



SPINAL CORD

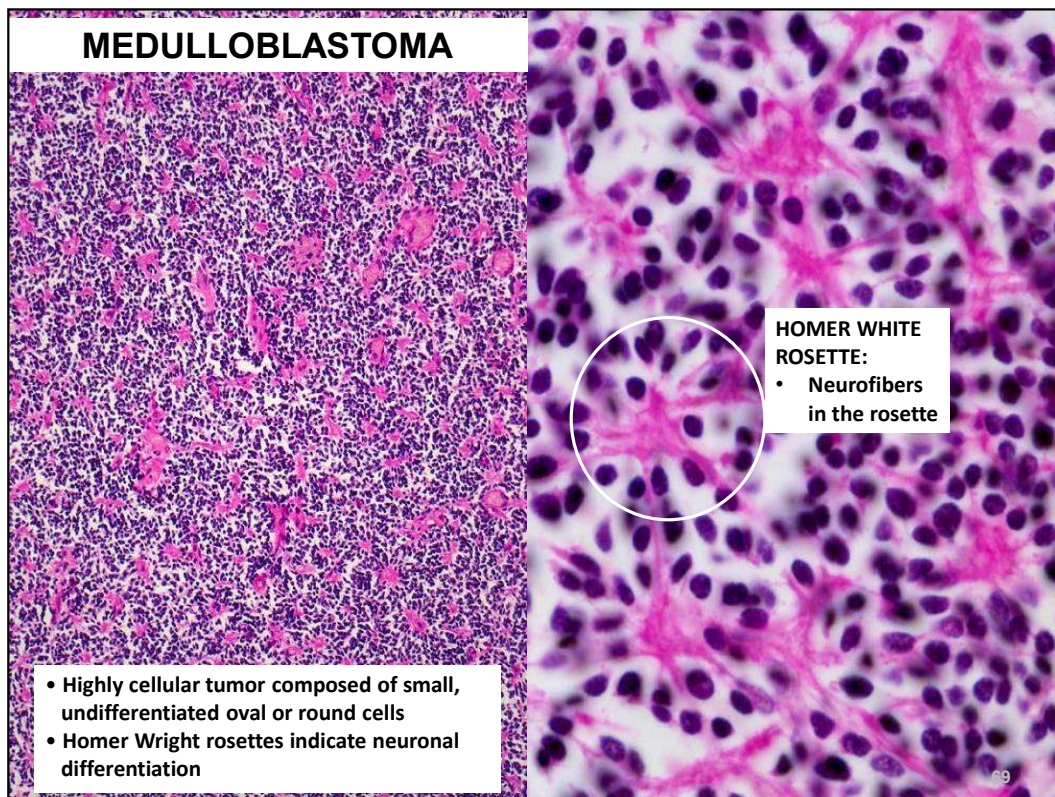


CAUDA EQUINA

Spinal cord is encased in tumor, which also glues together the nerve roots of cauda equina

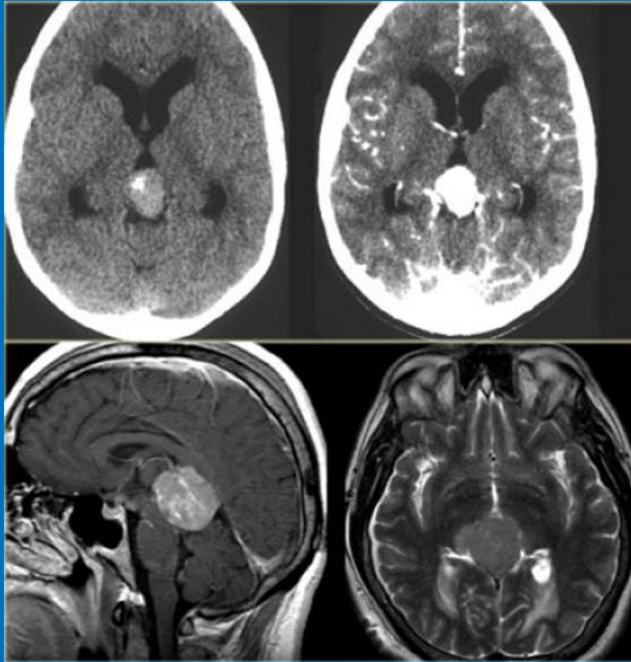
68

7-year-old boy. Diagnosis: medulloblastoma. **Description:** Blue cell-rich tumor located between the cerebellar cortex (CC) and the brainstem. **Clinical findings:** Headache and vomiting in the morning. **Stain:** HE.



7 year old boy. Diagnosis: medulloblastoma. **Description:** Blue highly cellular tumor composed of diffuse masses of small, undifferentiated oval or round cells. Some medulloblastomas show neuronal, glial and other differentiation. **Neuronal differentiation is manifested by neuropil and rosette formation. Rosettes** are groups of tumor cells arranged in a circle around a fibrillary center. Mature neurons may be found in medulloblastomas infrequently. **Glial differentiation is shown by GFAP-positive cells.** Focal differentiation into neuronal oligodendroglial or ependymal cells may be present. Rarely differentiation into striated muscle cells (medulloblastoma) and melanin-producing cells is noted. **Desmoplastic medulloblastoma** is a variant of medulloblastoma with a firm consistency and a collagenous stroma. Desmoplastic medulloblastomas tend to be located superficially and are circumscribed, such that they can be shelled out. The fibroblastic reaction occurs when the tumor extends into the subarachnoid space. Desmoplastic medulloblastoma is more frequent in the cerebellar hemispheres and in older patients. **Clinical findings:** Headache and vomiting in the morning. **Stain:** HE.

MIDLINE TUMOR IN THE PINEAL REGION WITH CALCIFICATIONS – MOST LIKELY DX: TERATOMA



GERM CELL TUMORS

TERATOMA
GERMINOMA
CRANIOPHARYNGIOMA

- In the US primary intracranial germ cell tumors account for only 2-5% of all CNS malignancies, and just over half of these are germinomas.
- Internationally primary intracranial germ cell tumors are more common
 - Account for up to 10% of all CNS malignancies in Asia, especially in Japan

There is a tumor located in the pineal region. The tumor contains calcifications. There is homogeneous enhancement, which is common for a tumor in the pineal region. Based on the age of the patient, the location and the tumor characteristics, this is most likely a teratoma

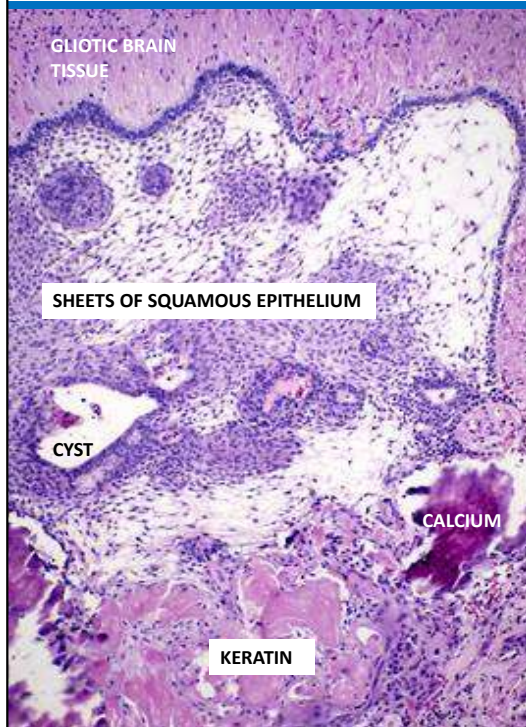
CRANIOPHARYNGIOMA



- Benign, slowly growing tumor (WHO Grade I)
- Arises from the epithelial remnants of Rathke's pouch trapped in the pituitary stalk
- Most often occurs in children and adolescents
- Cysts, calcification and the suprasellar location are the criteria for the radiologic diagnosis
- Can be locally destructive and cause a strong glial reaction in the surrounding brain
- Damages anterior pituitary and hypothalamus causing endocrine abnormalities
- Compresses the optic chiasm causing visual disturbances
- Blocks the 3rd ventricle causing hydrocephalus
- Difficult to resect completely; residual tumor is found in up to 40% of cases
- Other common suprasellar tumors are pilocytic astrocytoma and germ cell tumors

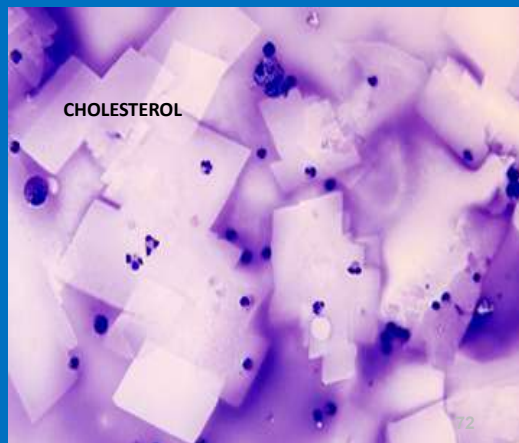
Cysts, calcification, and the suprasellar location are the criteria for the radiological diagnosis of craniopharyngiomas. Other common suprasellar tumors are pilocytic astrocytoma and germ cell tumors.

CRANIOPHARYNGIOMA

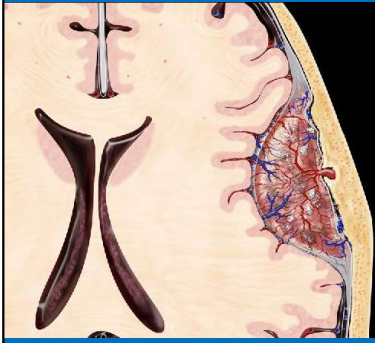


HISTOLOGY:

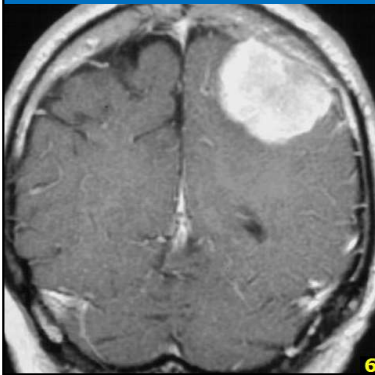
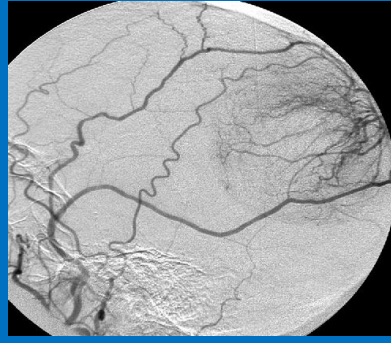
- Glial reaction in the surrounding brain
- Solid and cystic areas
- Sheets of squamous epithelial cells and keratin in loose connective tissue stroma
- Islands of keratin may calcify
- Cholesterol crystals may form in the cystic areas



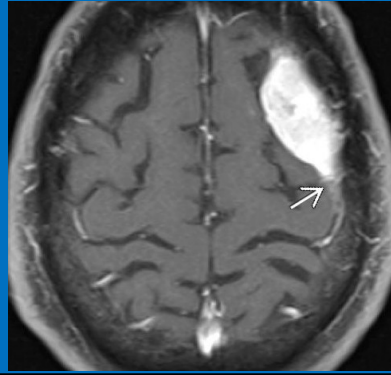
MENINGIOMA



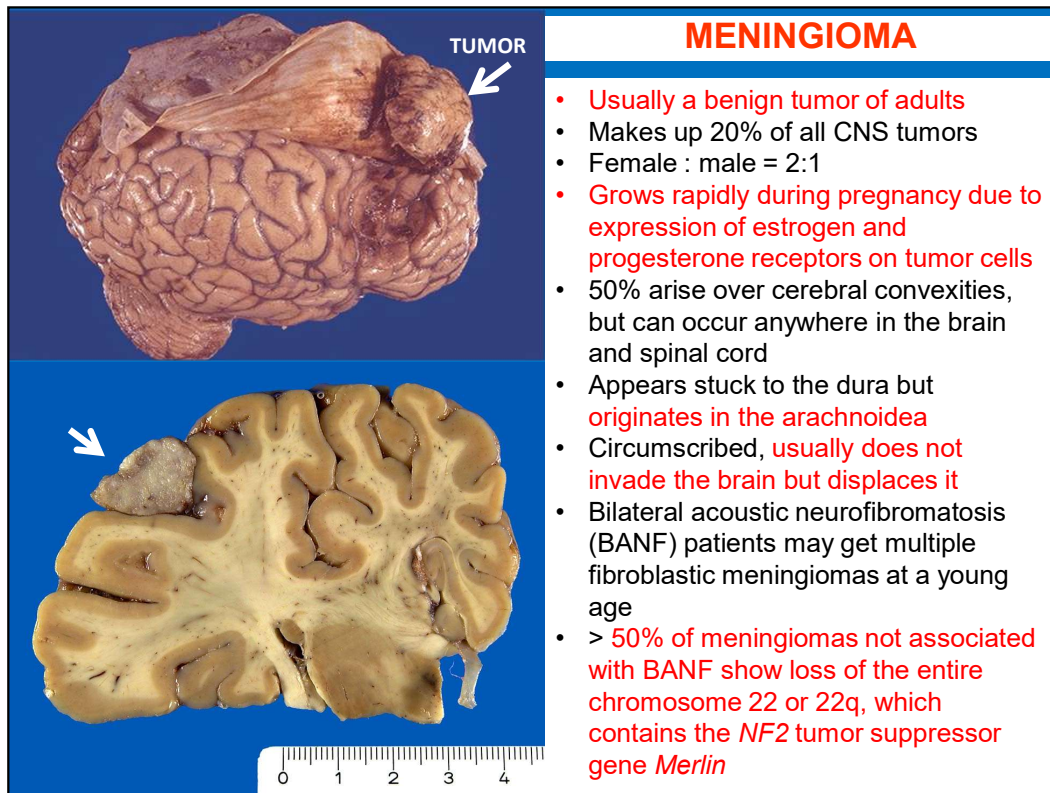
- Extra-axial mass
- Blood supply from external carotid artery



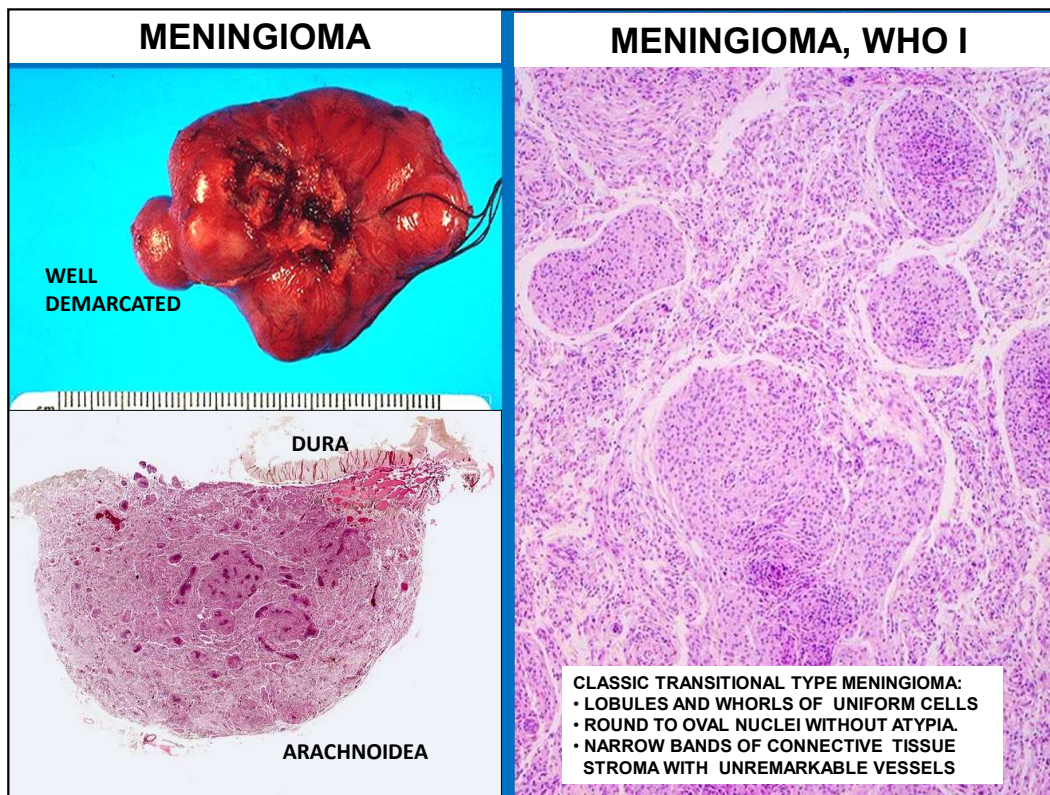
Note the "tails" = thickening of the meninges at the edges of the tumor mass



Meningioma Extra-axial mass. Angiogram of external carotid artery demonstrating enhancement



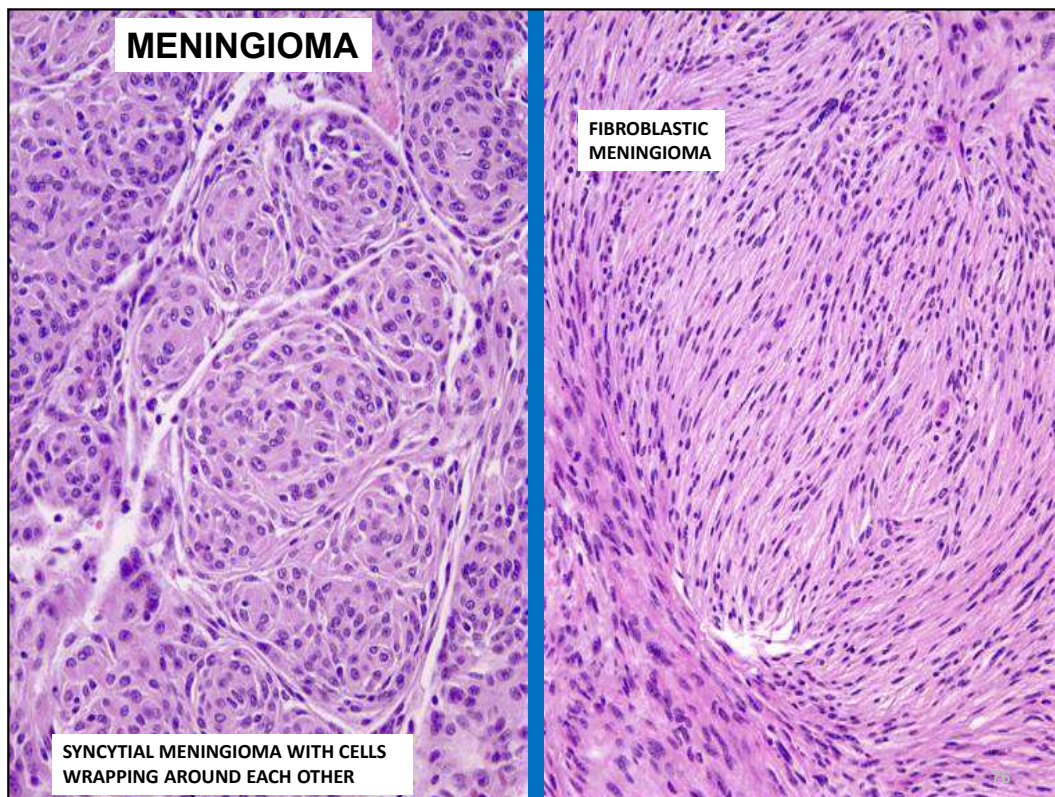
The meningioma beneath the dura has compressed the underlying cerebral hemisphere. Meningiomas can rarely be more aggressive and invade.



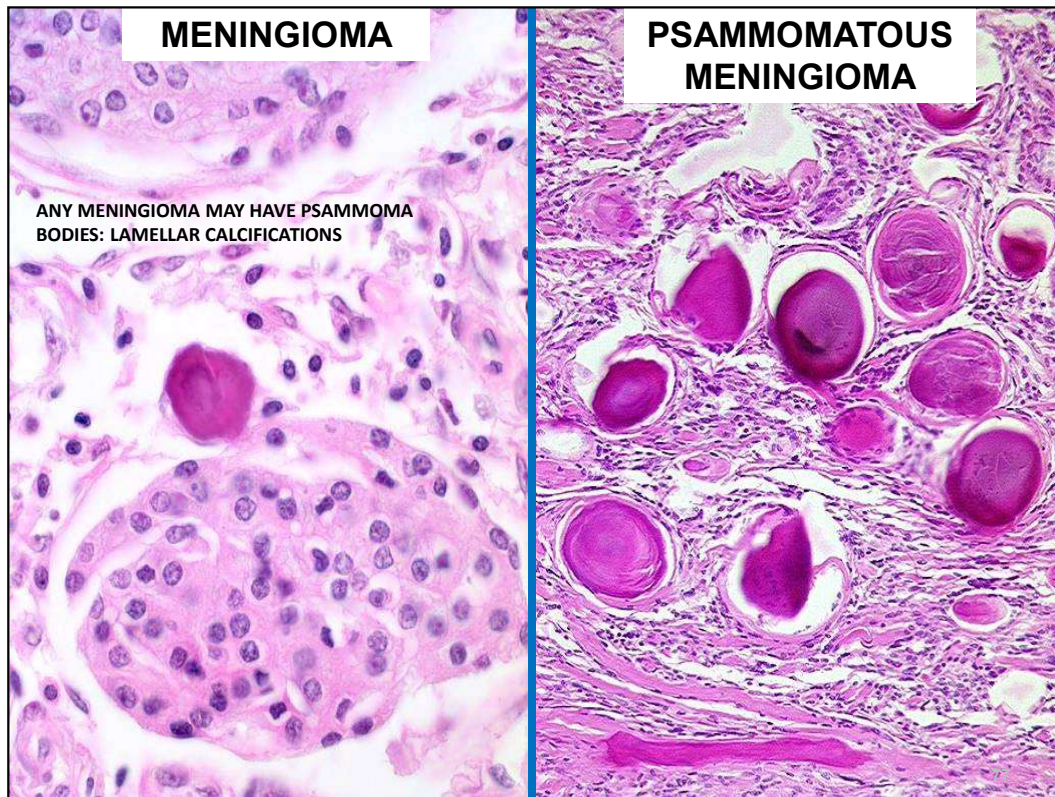
Left: Resected meningioma. **Right:** Resected meningioma, low power.

Clinical findings: 59 y o man had suffered several weeks of a right-sided headache and then had a first time epileptic seizure. He had a CT scan of the brain, which showed a 3 cm tumor over the right cerebral convexity.

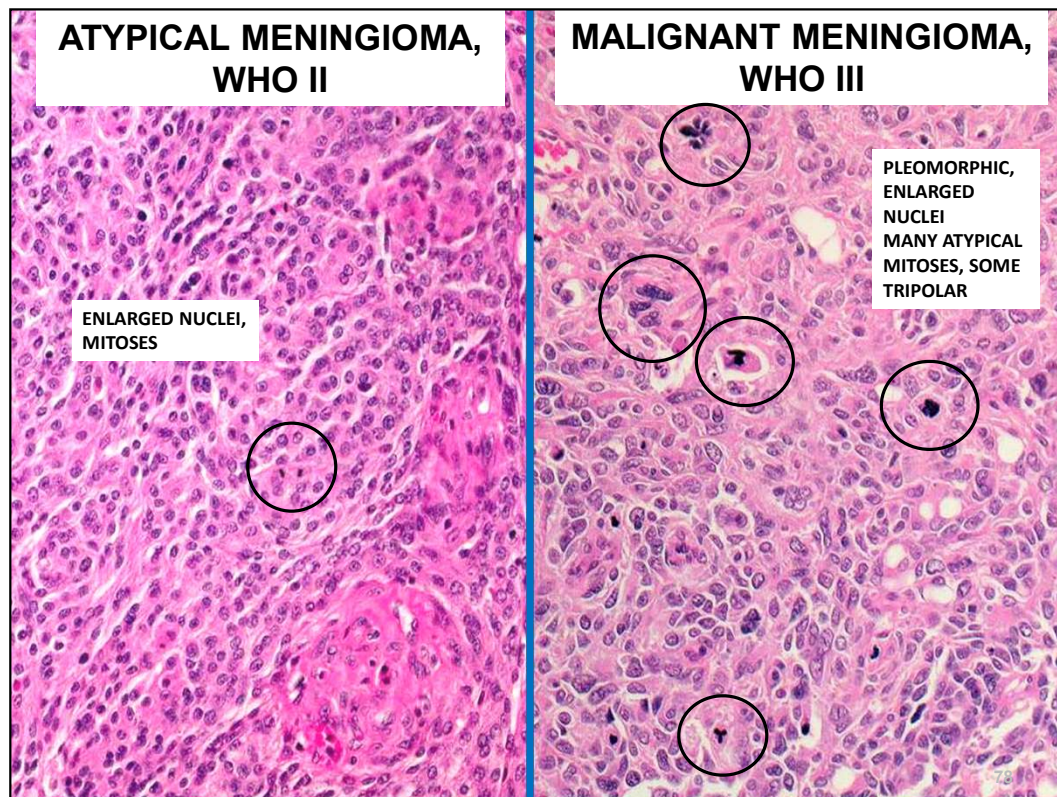
Diagnosis: Meningothelial meningioma (WHO I). **Description:** Sharply demarcated tumor attached to dura with classic features of meningioma. The tumor is composed of uniform round cells that form lobules and whorls. Between tumor cells are unremarkable blood vessels and narrow bands of connective tissue. **Stain:** HE



Syncytia: meningotheial cells wrap around each other

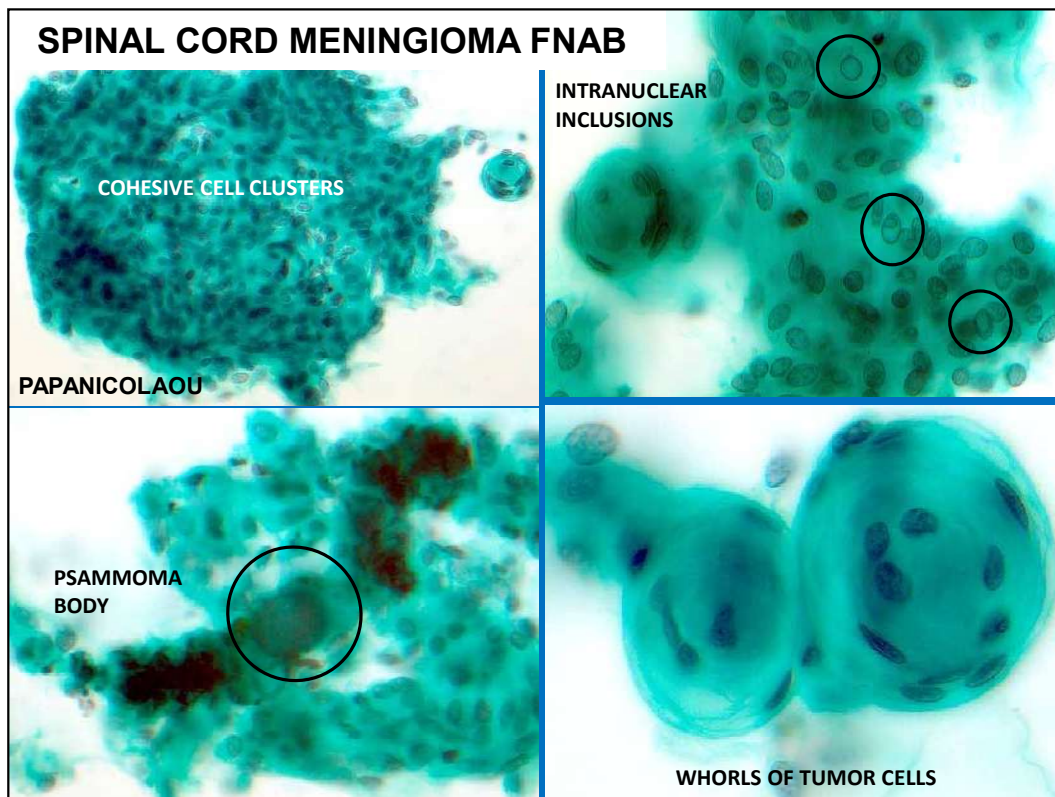


Left: 59 y o man had suffered several weeks of a right-sided headache and then had a first time epileptic seizure. The patient had a CT scan of the brain, which showed a 3 cm tumor over the right cerebral convexity. **Diagnosis:** Meningothelial meningioma (WHO I). **Description:** Sharply demarcated tumor attached to dura with classic features of meningioma. The tumor is composed of uniform round cells that form lobules and whorls. Between tumor cells are unremarkable blood vessels and narrow bands of connective tissue. Occasional psammoma bodies are present. **Stain:** HE. **Right:** A different patient with **psammomatous meningioma** with numerous psammoma bodies.



LEFT: 68 yo man. Diagnosis: Atypical meningioma (WHO Grade II). **Description:** Cell-rich tumor with many mitoses. **Additional findings:** The tumor infiltrated into falx cerebri and into vessels. **Clinical findings:** Recurring atypical meningioma on both sides of falx cerebri. **Comment:** Atypical meningiomas have increased number of mitoses (4 or more per 10 high power fields) or 3 or more of the following features: increased cellularity, small cells with increased nucleus to cytoplasmic ratio, focal or geographic necroses, absent or only focally apparent characteristic growth pattern. **Stain:** HE.

RIGHT: 68 y o woman. Diagnosis: Anaplastic malignant meningioma (WHO Grade III). **Description:** Highly cellular tumor with many severely atypical cells and high mitotic count (over 20 per 10 high power fields). There are many atypical mitoses. Many tumor cells contain intranuclear inclusions. **Additional findings:** The tumor was located at the cerebello-pontine angle and infiltrated into cerebellar parenchyma. **Comment:** Anaplastic meningiomas display histologic features of malignancy and may resemble carcinomas or sarcomas. Median survival is <2 years. **Stain:** HE.

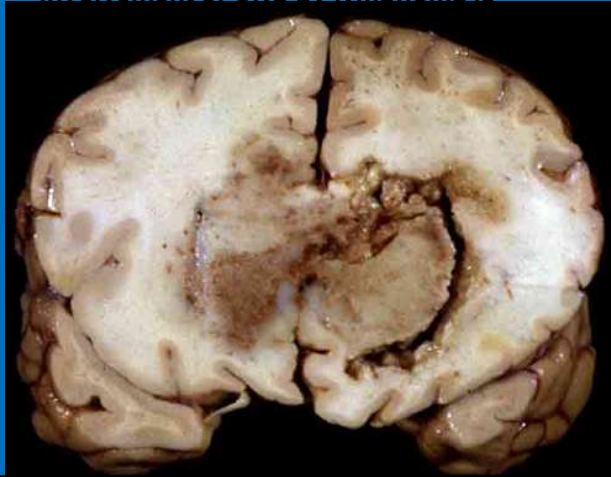
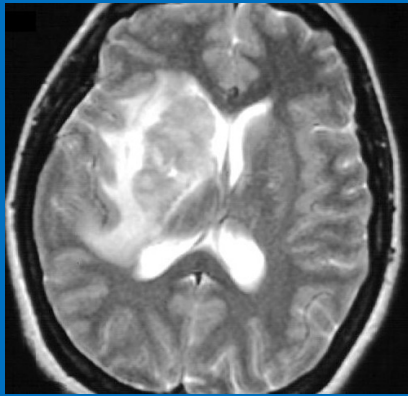
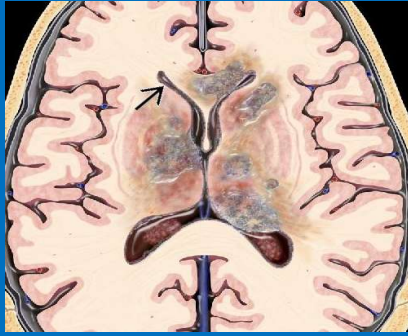


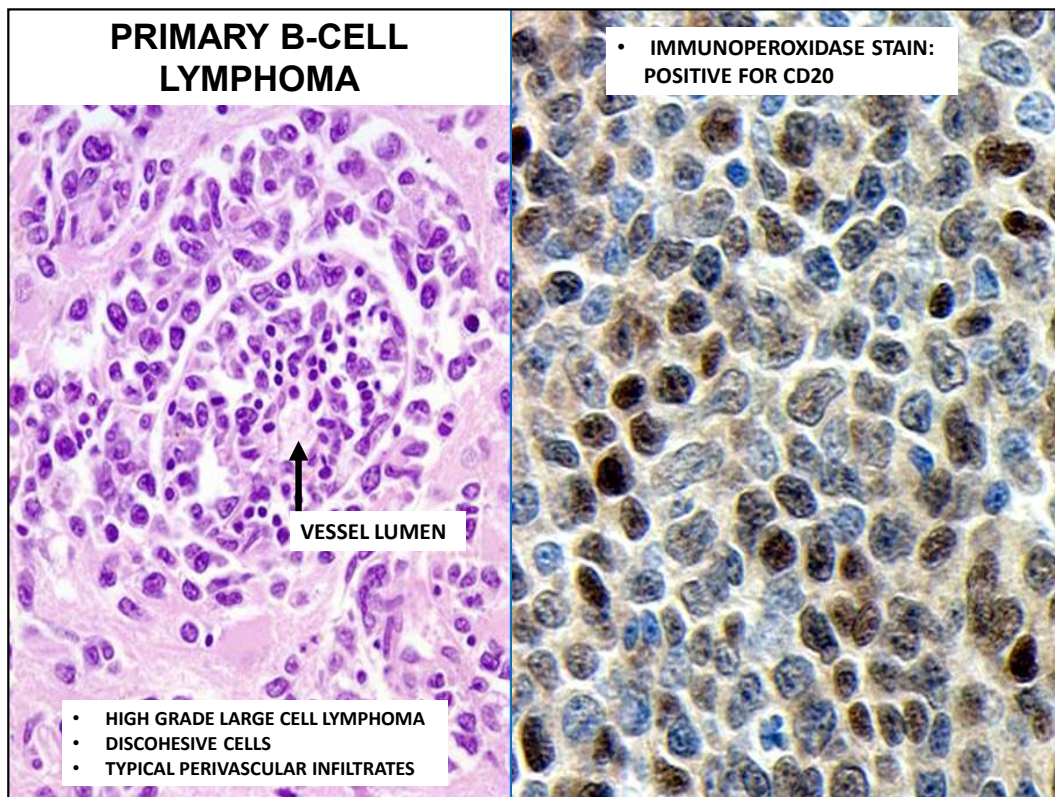
Clinical information: 39 y o woman with hemiparesis and a mass in spinal cord at the level of T12. **Diagnosis:** Meningioma. **Description:** Cohesive cell clusters composed of uniform cells with abundant cytoplasm and round to oval nuclei. Many calcified psammoma bodies are present. Fine needle aspiration biopsy. **Stain:** Papanicolaou.

CNS LYMPHOMA

- **Most patients with primary CNS lymphoma have AIDS or are immunosuppressed**
- Origin: resident CNS lymphoid cells
- Grossly destructive like GBM
- Besides primary CNS lymphomas, metastatic

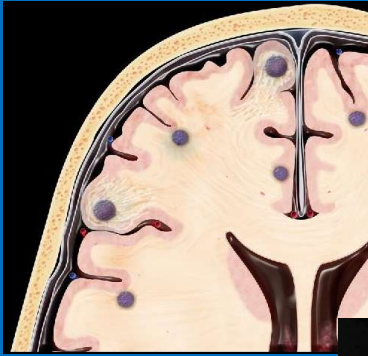
involvement of CNS by lymphoma is





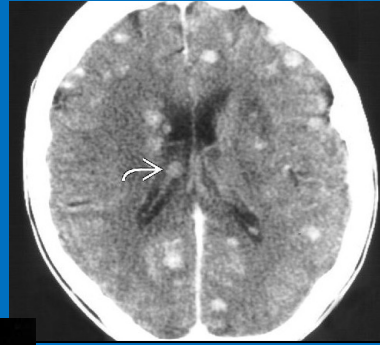
Immunohistochemical studies revealed the tumor cells to be negative for epithelial membrane antigen (EMA), melanocyte marker (HMB-45) and glial fibrillary acid protein (GFAP) and **positive for leucocyte common antigen (LCA/CD45)**. The majority of the cells were also **positive for B-cell antigen (CD20)**.

METASTATIC TUMORS TO THE CNS

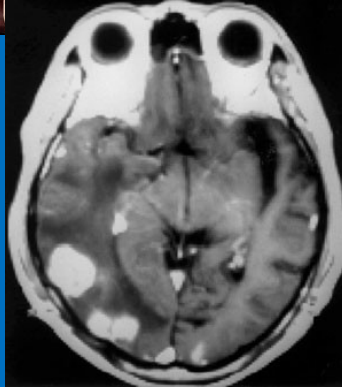


Common location:

- Junction of gray and white matter



- Metastasis from another primary site is the most common tumor in the brain in adults
- Multiple lesions suggest metastatic cancer
- Up to 1/3 are from an unknown primary
- Work-up should include CT of chest, abdomen and pelvis



- Palliative treatment:
 - Glucocorticoids,
 - Anticonvulsants
 - Radiation therapy
 - Surgery, if a single accessible lesion only present
 - Chemotherapy directed to the primary neoplasm

METASTATIC TUMORS TO CNS

TUMOR INFILTRATES:

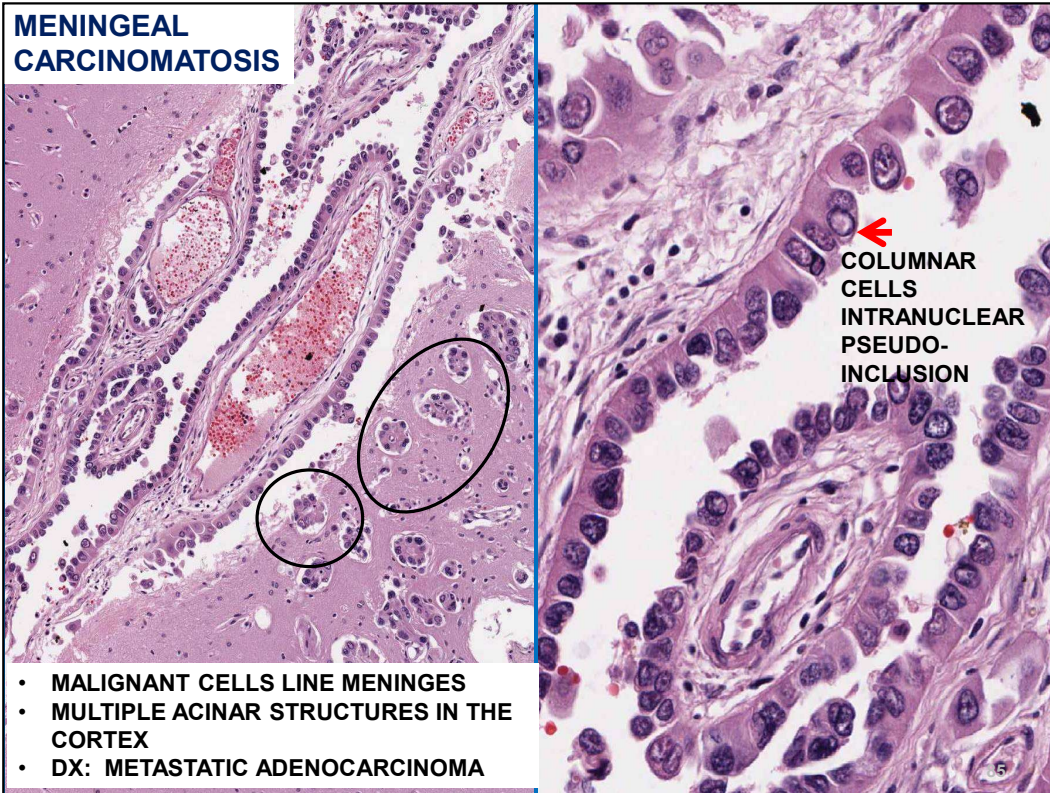
- Brain metastases are found at autopsy in 14-37% of patients with malignant tumors
- Often multiple
- Any tumor can metastasize to CNS
- CNS metastases are most often from lung, (particularly small cell lung cancer) and breast carcinomas
- Melanoma has the highest frequency
- Lymphoid malignancies (particularly ALL) metastasize often to the CNS
- Meningeal carcinomatosis is seen in 4-8% of metastatic tumors to the CNS
- CSF in meningeal carcinomatosis: high protein, low glucose, a few lymphocytes and tumor cells

METASTATIC TUMORS TO CNS

METASTATIC TUMORS TO CNS MAY CAUSE PARANEOPLASTIC SYNDROMES:

- **Subacute cerebellar degeneration** resulting in ataxia with destruction of Purkinje cells, gliosis and inflammation
- **Limbic encephalitis** resulting in subacute dementia with perivascular inflammation, microglial nodules, neuronal loss and gliosis located in the medial temporal lobe
- **Subacute sensory neuropathy** leading to altered pain sensation with loss of sensory neurons from dorsal root ganglia and inflammation
- **Syndrome of rapid-onset psychosis, catatonia, epilepsy and coma** associated with ovarian teratomas and antibodies against the N-methyl-D-aspartate (NMDA) receptor

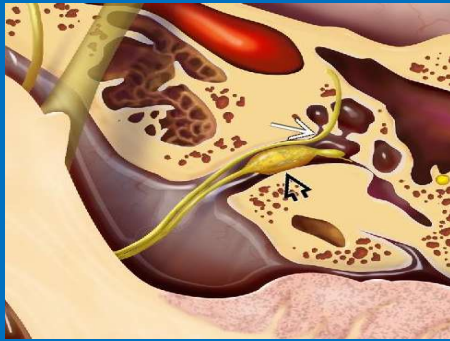
MENINGEAL CARCINOMATOSIS



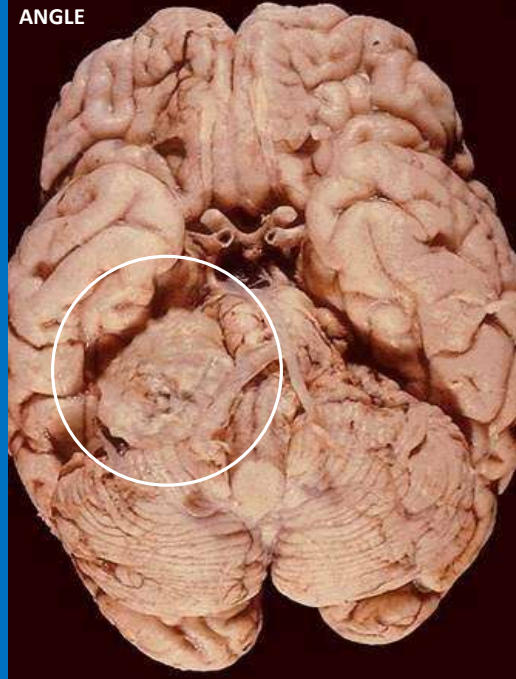
- MALIGNANT CELLS LINE MENINGES
- MULTIPLE ACINAR STRUCTURES IN THE CORTEX
- DX: METASTATIC ADENOCARCINOMA

←
COLUMNAR
CELLS
INTRANUCLEAR
PSEUDO-
INCLUSION

ACOUSTIC SCHWANNOMA



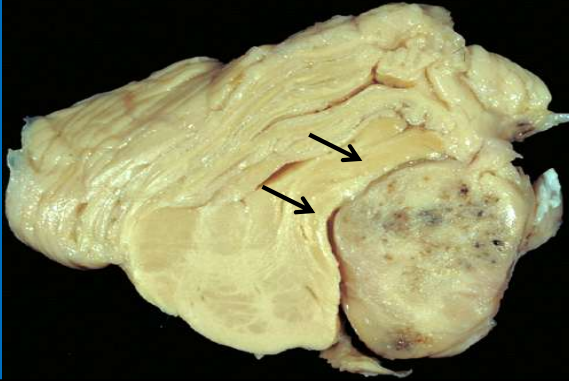
TYPICAL LOCATION IN THE CEREBELLO-PONTINE ANGLE



Acoustic Schwannoma 5mm mass

NEUROMA (SCHWANNOMA)

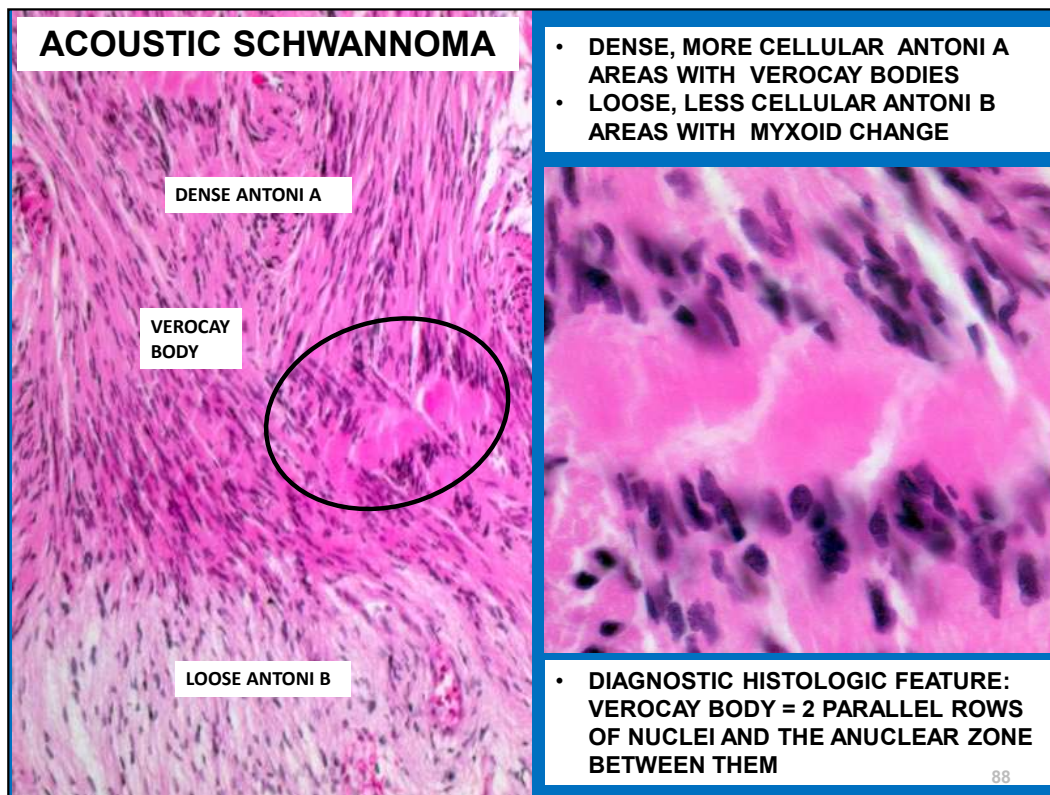
- Arises in cranial and spinal nerve roots
- 90% are in the 8th cranial nerve in the cerebello-pontine angle (= acoustic neuroma)
- Extra-axial, benign and slow-growing
- Well-circumscribed and encapsulated
- Symptoms are due to compression
- Can be usually completely removed
- "Fish flesh" gross appearance
- May be hemorrhagic



FISH
FLESH

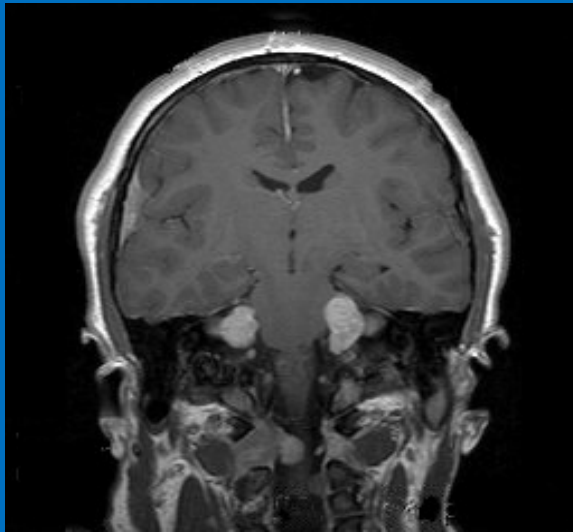


87

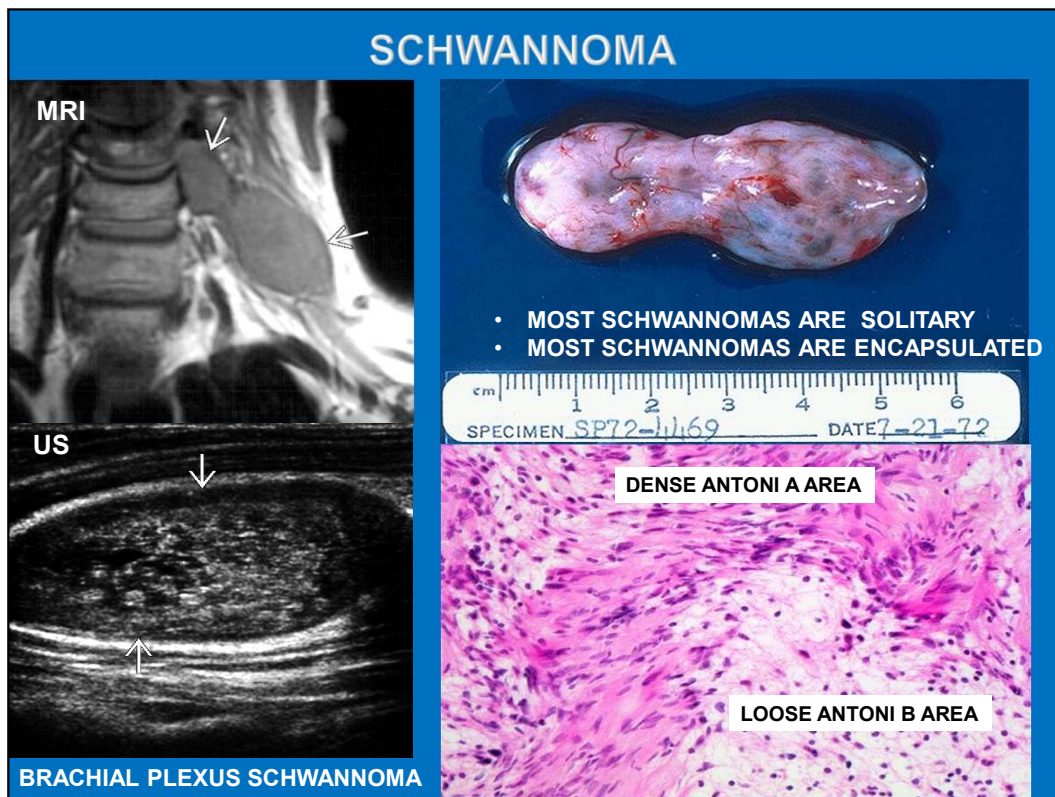


52 y o man. Diagnosis: Acoustic neuroma = Schwannoma. **Description:** Cellular Antoni A-areas with Verocay bodies (row of palisading = parallel nuclei) and less cellular Antoni B areas with myxoid change. **Stain:** HE.

NEUROFIBROMATOSIS 2 (NF2)



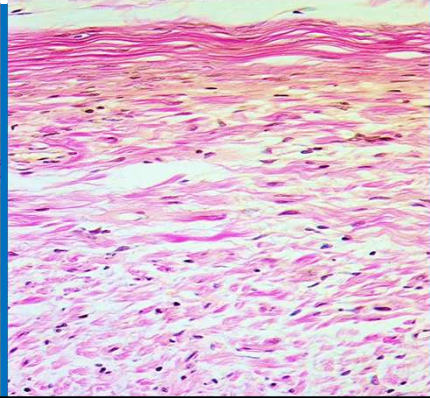
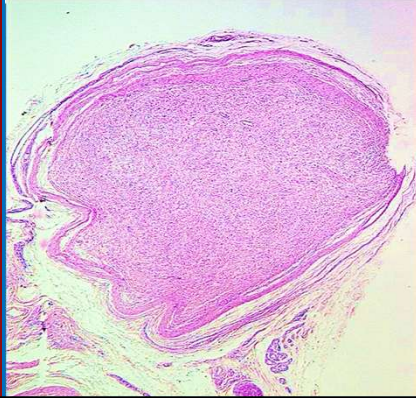
- Bilateral acoustic neuromas (BANF) in NF2
- The product of *NF2* gene merlin inhibits cell proliferation and cell cycle progression by repressing cyclin d1 promoter activity



MRI and Ultrasound of a brachial plexus schwannoma

NEUROFIBROMA

- Peripheral nerve tumor
- Not encapsulated
- Proliferation of a mixture of Schwann cells and fibroblasts causes fusiform enlargement of the nerve
- Schwann cells and fibroblasts are arranged loosely in a wavy pattern
- Multiple neurofibromas involving long segments of peripheral nerves create plexiform structures characteristic of neurofibromatosis 1 (NF 1)

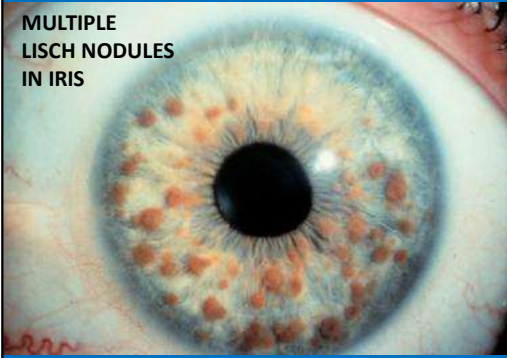


NEUROFIBROMATOSIS NF1 (VON RECKLINGHAUSEN)

- Autosomal dominant disorder caused by mutations in the tumor suppressor **neurofibromin**
- Gene locus in the long arm of chromosome 17 (17q).
- Neurofibromin is a negative regulator of the oncoprotein RAS
- Loss of neurofibromin function is a main feature of NF1-associated tumors.
- The single copy of normal *NF1* allele is mutated or silenced in tumors arising in the setting of NF1.
- Patients develop localized cutaneous neurofibromas, plexiform neurofibromas (these have malignant potential), diffuse neurofibromas, malignant peripheral nerve sheath tumors, "optic gliomas," and other glial tumors.
- Patients with NF1 also have other problems:
 - learning disabilities,
 - Seizures
 - skeletal abnormalities
 - vascular abnormalities including arterial stenoses,
 - pigmented nodules of the iris (Lisch nodules)
 - pigmented skin lesions e.g. axillary freckling and café-au-lait spots.

NEUROFIBROMATOSIS NF1

MULTIPLE
LISCH NODULES
IN IRIS

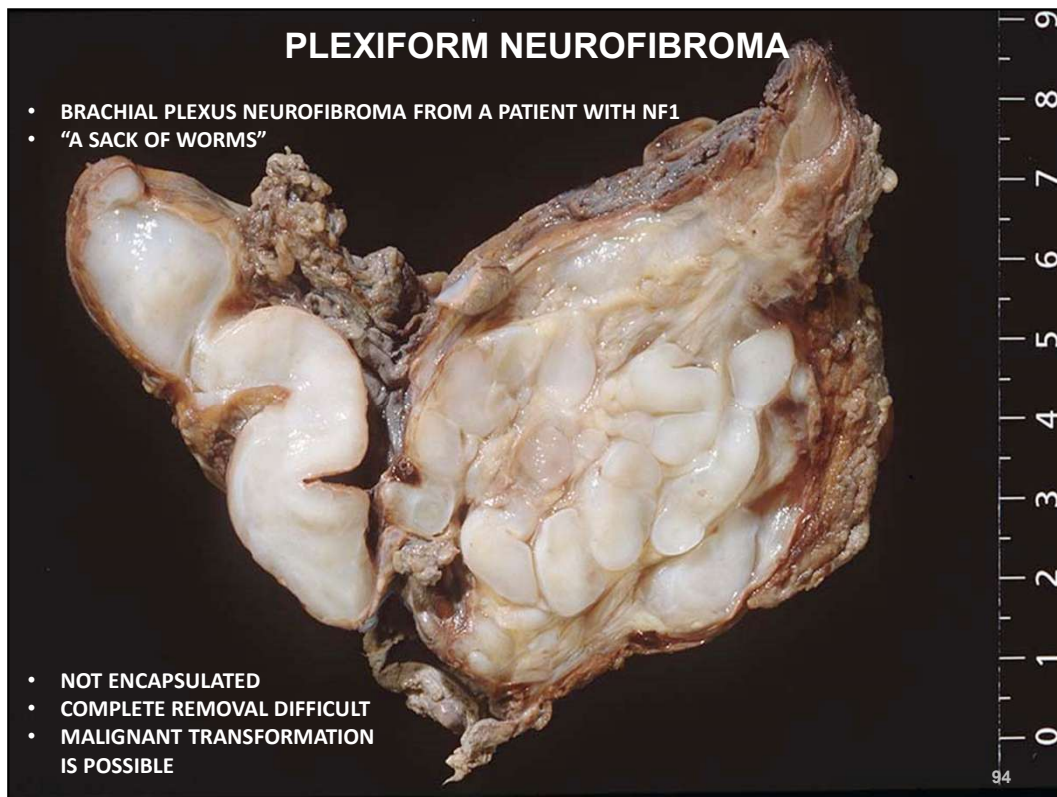


NEUROFIBROMATOSIS TYPE 1 (NF1)
WITH MULTIPLE NEUROFIBROMAS
DEVELOPING IN A CAFÉ AU LAIT SPOT



CAFÉ AU LAIT SPOTS



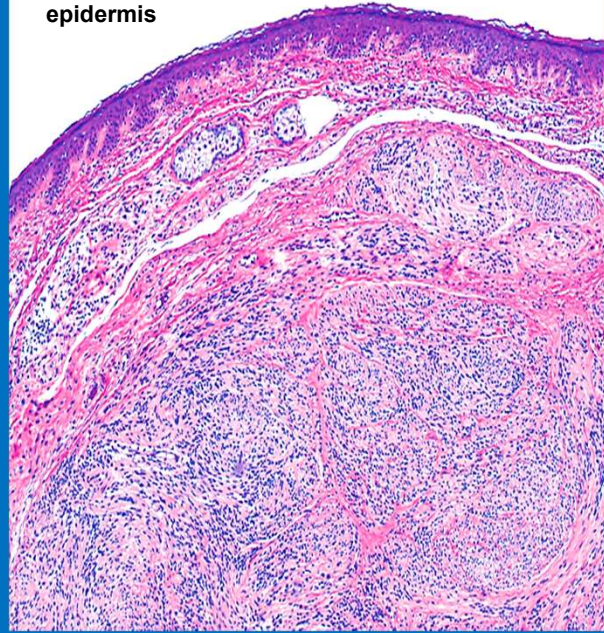


17 y o female. Diagnosis: Plexiform neurofibroma of the brachial plexus.
Description: Worm-like tumor masses. **Clinical findings:** Neurofibromatosis type 1. The patient has a mutation in the gene located in the long arm of chromosome 17 (17Q) encoding the tumor suppressor neurofibromin.

PALISADED ENCAPSULATED NEUROMA AKA SOLITARY CIRCUMSCRIBED NEUROMA

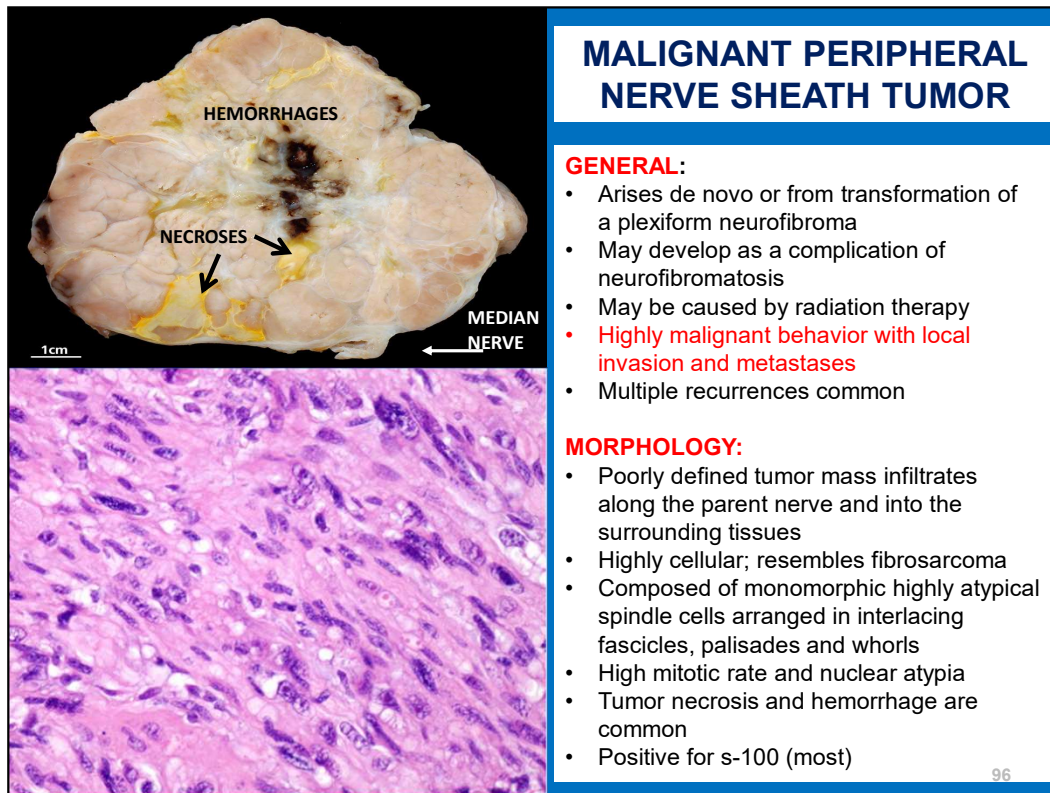


Covered by intact
epidermis



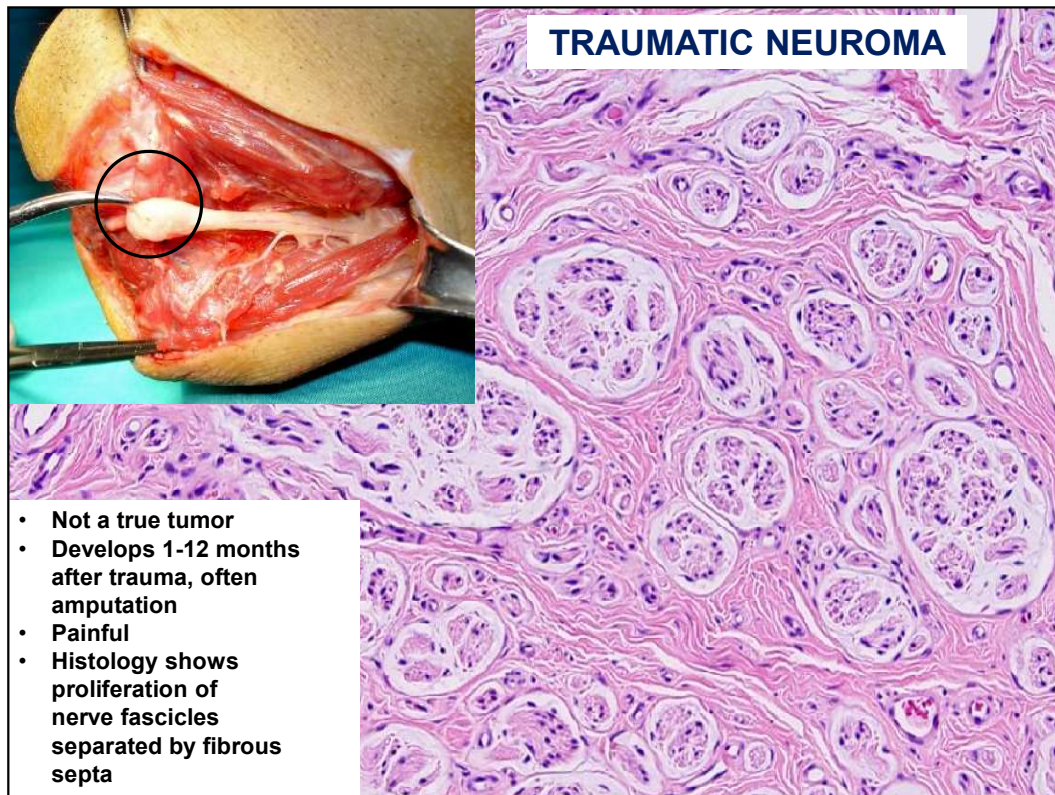
- Small, solitary skin papule
- Asymptomatic
- Typically in the face of a middle-aged person
- Composed of proliferation of Schwann cells and axons separated by benign connective tissue
- Encapsulated
- Not associated with neurofibromatosis

Courtesy of Mark R. Wick, MD



45 yof. Diagnosis: Malignant peripheral nerve sheath tumor. **Description:** Lobulated tumor with focal necrosis. **Clinical findings:** Tumor in the median nerve.

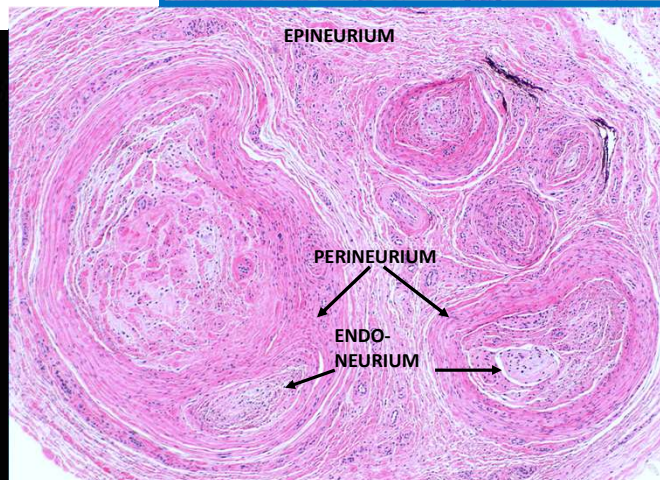
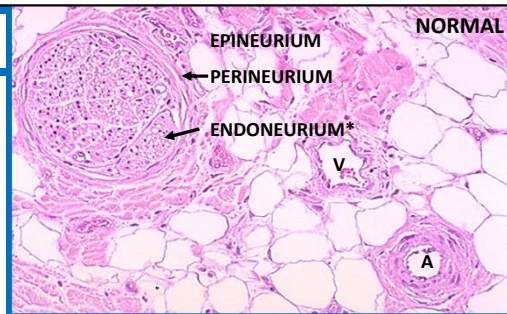
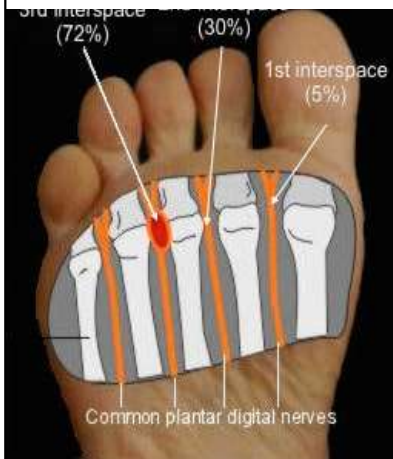
Genetics:



Courtesy of Jian-Hua Qiao

MORTON'S NEUROMA

- Aka Morton's metatarsalgia, plantar neuroma, intermetatarsal neuroma, plantar interdigital neuroma
- Not a true tumor but a perineural fibrous proliferation
- Tight shoes (high heels) and gait abnormalities cause trauma and entrapment of common plantar digital nerves at the bifurcation point



* Endoneurium is a layer of delicate connective tissue that encloses the myelin sheath of a nerve fiber and is made up of loose, supportive endoneurial connective tissue.

